RECOGNISING BEFORE REACTING: TEMPORAL SIGNATURES OF TRANSACTIONAL GAMES

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Abstract. We formalise a source-verified corpus of seven transactional games — six classical Berne games from *Games People Play* plus one project-derived variant — as finite-state interaction systems and study the Karpman projection π_K that maps each game to its Drama-Triangle role-configuration trace. Two structural findings emerge. First, π_K is not injective on the corpus: three source-distinct games (Why Don't You Yes But, If It Weren't For You, See What You Made Me Do) induce an identical role trace. Second, the map from a game to the intervention sub-type required by its source-prescribed *antithesis* does not factor through the Karpman switch permutation: WDYB and SWYMD induce the same switch permutation but require distinct intervention types. We package the switch as an S_3-action on the role space and give a machine-checked witness that the two switch generators observed in the corpus generate all of S_3. The mathematical machinery for locating the antithesis executably — diagnosability delay t_{obs} [SSL+95], coprognosability reaction bound t_{redir} [KT10], and signed antithesis window $W = t_{\text{redir}} - t_{\text{obs}}$ — is imported from discrete-event systems and runtime verification; no novelty is claimed for the imported apparatus. The classification of the corpus by W-sign into REACTIVE, PRECOMMITTED, and KNIFE-EDGE regimes depends on the alternative-class supplied to the observer and is exhibited jointly with its parameterisation.

1. Introduction

Eric Berne's *Games People Play* [Ber64] catalogues a family of stereotyped social interactions — *transactional games* — in which two or more participants execute a stereotyped exchange terminating in a predictable emotional *payoff*. Berne prescribes, per game, an *antithesis*: an interaction move by one participant that prevents the game from completing. Stephen Karpman's *Drama Triangle* [Kar68] projects the participants of a game onto a role space $\{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}\}$ (Victim, Persecutor, Rescuer) and observes that drama analysis reveals repeated switches of role and location. Berne and Karpman together supply a rich source domain but leave open a formal question:

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When can a transactional game be observed early enough for its source-prescribed antithesis to remain executable?

The question is temporal (it asks about prefixes of an interaction), observational (it asks what is visible to whom), and interventional (it asks what actions are available). These are the natural coordinates of finite-trace temporal reasoning [BLS11] and discrete-event control [RW87]. The bridge is not mathematically deep: the two boundary quantities we need are already present in the DES literature as *diagnosability delay* [SSL$^+$95] and *reaction bound for coprognosis* [KT10]. What is new is the source-faithful application to a specific domain corpus, plus two structural findings that would not have surfaced without the domain projection.

Contributions. Ordered by novelty and formal strength:

(1) **Non-injectivity of the Karpman projection.** On the seven-game corpus, three source-distinct games (WDYB, IWFY, SWYMD) induce an identical role-configuration trace, so the Karpman projection π_K from Berne games to Drama-Triangle role traces is not injective (Proposition 9.1, Corollary 9.3).
(2) **Permutation-only classification refuted.** WDYB and SWYMD induce the same Karpman switch permutation but require distinct intervention sub-types; consequently the sub-type map does not factor through the switch permutation (Proposition 9.6).
(3) **Group-theoretic reframing of the Karpman switch.** We package role-switches as a group action of S_3 on the role space $\{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}\}$ (Definition 6.1) and give a machine-checked witness that the two switch generators observed in the corpus generate all of S_3 (Proposition 6.2).
(4) **Source-verified machine-checked corpus.** A typed finite-trace formalisation of six canonical Berne games plus one project-derived variant (Section 7), with explicit separation of source-derived objects, programme abstractions, and code representations, and per-passage provenance markers (P0 primary-verified vs S1 secondary-attested).
(5) **Alt-class fragility of REACTIVE / PRE-COMMITTED classification.** The W-sign classification of the corpus is not invariant under the alternative-class choice: NIGYSOB and KICK_ME are PRE-COMMITTED under $\mathfrak{A}_{\text{nongame}}$ and KNIFE-EDGE under $\mathfrak{A}_{\text{dataset}}$ (Remark 8.1). The classification is a joint property of the game and the observer's alt-class, not of the game alone.

The remaining formal apparatus — observational-identifiability frontier t_{obs} (Definition 5.3), redirection frontier t_{redir} (Definition 5.10), signed antithesis window $W = t_{\text{redir}} - t_{\text{obs}}$ (Definition 5.11) — is a domain specialisation of the diagnosability and coprognosability constructions of [SSL$^+$95, KT10]; no novelty is claimed for the imported machinery (Section 14).

Limits. This paper does not clinically validate Transactional Analysis; role projections onto the Drama Triangle are modelling choices, not derived from Karpman's own text (see Section 2); every result is a corpus-level statement, not a universal claim about transactional dynamics; the intervention classes are formal, not clinical prescriptions. Section 16 enumerates these limits fully.

2. Transactional Analysis as source domain

We separate three source layers throughout: what Berne 1964 [Ber64] and Berne 1972 [Ber72] actually state; what Karpman 1968 [Kar68] adds; and what this programme introduces as formal abstraction.

Berne 1958. The earliest published description of the game Why Don't You Yes But (WDYB) appears in Berne 1958 [Ber58], six years before *Games People Play*. Berne describes the Hyacinth vignette (pp. 740–741):

> *Hyacinth had told her story for the purpose of setting in motion the commonest of all the games which can be observed in groups: the game of "Why don't you ... Yes but ... "* [Ber58].

Berne 1958 also prints the SCHLEMIEL antithesis verbatim (p. 740, [Ber58]: "*You can break things and spill things all you like; but please don't say 'I'm sorry!' "*), a PRIMARY-VERIFIED (P0) anchor that does not require access to [Ber64].

Berne 1964. *Games People Play* [Ber64] catalogues the game family, including the seven games this paper models. The *Antithesis* paragraphs Berne prescribes per game are cited throughout Section 8; the per-passage provenance level (primary content verified against [Ber64] vs. secondary attestation only) is recorded in Appendix A. Berne also states in his opening chapter on games that a game admits a *minimum number of moves essential to its progress*, expressible as a *protocol* — a historical anchor for the finite-round formalisation of Definition 4.1, without implying Berne developed a formal protocol semantics.

Berne 1972. Formula G — the six-symbol schema $C + G = R \to S \to X \to P$ (Con + Gimmick = Response → Switch → Crossup → Payoff) — is a Berne 1972 construct, per Karpman's own 2019 attestation [Kar19]:

> *Berne (1972) never created a script formula as he did a game formula* [Kar19].

Berne 1964 introduces the game structure (con, gimmick, switch, payoff) without the acronym-string; the acronym-string form is a Berne 1972 construct [Ber72], whose attribution to Berne 1972 (as opposed to Berne 1964) is corroborated by Karpman's verbatim 2019 attestation above.

Karpman 1968. The Drama Triangle introduces three action roles Persecutor, Rescuer, Victim, distinguished from identity roles, and states that "drama can be analyzed as switches in role and location on a time continuum" [Kar68] . *Karpman 1968 does not name Why Don't You Yes But or Now I've Got You You Son of a Bitch*; the sole worked example is "I'm Only Trying to Help You". Consequently, projections of role labels $\{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}\}$ onto the Berne 1964 games listed in Section 7 are *project-original* projections of Berne's transactional structure onto Karpman's later role space, not Karpman's own analyses.

Karpman 2019. Extends the framework with fifteen further triangle variants [Kar19] without changing the 1968 definitions of Persecutor / Rescuer / Victim.

Notation and role-name discipline. Throughout, we use typed role names Vic, Per, Res (Victim, Persecutor, Rescuer) rather than the unqualified P/R/V letters. This avoids collision with the Formula G symbols P (Payoff), R (Response), S (Switch), and with Berne ego states Parent, Adult, Child (which we do not need mathematically in this paper but which use overlapping initials in the source literature).

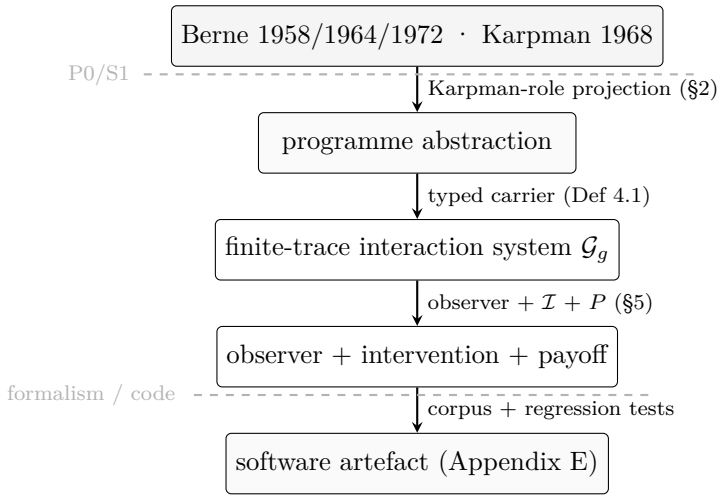


FIGURE 1. Semantic layers. Every downward arrow is a modelling step whose provenance status is tracked in Appendix A. Source layer is Berne/Karpman verbatim; layers below are project abstractions.

3. Formal preliminaries

We work with finite alphabets, finite states, and finite prefixes of a potentially infinite trace. LTL fluents are interpreted over finite prefixes with the three-valued semantics of Bauer, Leucker, and Schallhart [BLS11]: a trace-prefix $\tau_{\leq k}$ produces one of the verdicts $\{\top, \bot, ?\}$ against a specification φ. We use the safety–cosafety–liveness–coliveness classification of Alpern–Schneider as refined in [Bra26a]; the four classes are denoted $\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}$.

We adopt the following semantic-hygiene lemma:

Lemma 3.1 (Language class is language-intrinsic)**.** *For any LTL formula φ over an alphabet Σ and any observation model O, the class* $\mathrm{Class}(L(\varphi)) \in \{\mathsf{S}, \mathsf{C}, \mathsf{L}, \mathsf{K}\}$ *of the language $L(\varphi) \subseteq \Sigma^\omega$ is independent of O. The* pair-class $\mathrm{Class}(L(\varphi), O)$ *— the monitorability of the (formula, observation-model) pair — may differ from* $\mathrm{Class}(L(\varphi))$ *under a non-trivial O.*

Proof. The class of an ω-language is a topological invariant of the language regarded as a subset of Σ^ω [MP90, Bra26a]; no observation parameter enters the definition. Pair-class dependence on O is the substance of [Bra26d] Theorem `thm:obs-relative`, which we import here. □

Lemma 3.1 is stated to prevent the category error that an agent, by changing perspective, can change the class of a formula. An agent can only change which formula they *select*, or which observation model they use to monitor it; the language class of a fixed formula is intrinsic.

4. Temporal interaction model

We collect the ingredients of the corpus into a single carrier:

Definition 4.1 (Interaction system)**.** An *interaction system* for a Berne game g over agent set A is a tuple

$$\mathcal{G}_g \;=\; (Q_g,\, E_g^c,\, E_g^u,\, \delta_g,\, \Phi_g,\, P_g),$$

where

- $Q_g \subseteq \{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}, \mathsf{CI}\}^A$ is the state space of role configurations (one role per agent per tick, with CI denoting the non-drama role).
- E_g^c is the *controllable* event alphabet: $E_g^c = A \times \Delta$ where $\Delta \subseteq \{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}, \mathsf{CI}\}^2 \cup \{\mathsf{stutter}\}$ enumerates all single-agent role transitions plus stutter. An event $(a, r \to r') \in E_g^c$ is an intentional move by agent $a \in A$.
- E_g^u is the *uncontrollable* event alphabet, initialised as $\emptyset$ in the base model and unused across the current corpus (Section 7).
- $\delta_g : Q_g \times (E_g^c \cup E_g^u) \to Q_g$ is the (possibly non-deterministic in the presence of uncontrollable events) transition function. On E_g^c it is deterministic; on E_g^u it may be a relation.
- $\Phi_g \subseteq \mathrm{LTL}(AP_g)$ is a family of temporal specifications over game-fluents $AP_g = \{\mathsf{viol}, \mathsf{payoff}, \mathsf{switch}, \mathsf{exit}\}$, defined pointwise on Q_g or on pairs of consecutive states (for switch).
- P_g is the payoff condition, a subset of finite prefixes Q_g^* closed under extension (a payoff-locked prefix stays payoff-locked). In the base model $P_g = \{\tau : \mathsf{viol}$ holds and remains true$\}$.

Executions of $\mathcal{G}_g$ are finite sequences $h = q_0 e_0 q_1 e_1 \cdots q_n$ with $q_{i+1} \in \delta_g(q_i, e_i)$ for all $i < n$; $\mathrm{Hist}(\mathcal{G}_g)$ denotes the set of finite executions.

Remark 4.2 (Field elimination)*.* The prior formulation carried an O_g and a ρ_g field inside $\mathcal{G}_g$. They are removed here: observers are typed independently over $\mathrm{Hist}(\mathcal{G}_g)$ in Definition 5.1, and ρ_g is the identity on Q_g under our encoding ($Q_g \subseteq \{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}, \mathsf{CI}\}^A$) so it carries no information. The intervention class $\mathcal{I}$ is also removed as a per-game field; it is a paper-level parameter (Definition 5.4) that varies across experiments.

Definition 4.3 (History annotation)**.** Executions may be enriched with observation-time annotations without changing the system dynamics. An *annotation function* for $\mathcal{G}_g$ is a mapping

$$\mathrm{ann}_g : \mathrm{Hist}(\mathcal{G}_g) \to \Sigma_{\mathrm{ann}}^*,$$

where Σ_{ann} is the annotation alphabet. Standard annotations used in this paper: (i) $\mathrm{ann}^{\mathrm{conv}}$ maps each tick's role configuration to a fixed conversational-pattern label (*help-seeking*, *watcher-target*, *full-triangle*, …— see Section 5); (ii) $\mathrm{ann}_a^{\mathrm{sub}}$ maps each tick to agent a's substrate content. Annotations are DERIVED from h under a fixed rule; they do not add information outside h.

Fluents. From the corpus model, four Boolean fluents per state suffice:

$$\begin{aligned}
\mathsf{viol}(q) &= [\rho(q) \text{ contains both } \mathsf{Vic} \text{ and } \mathsf{Per}] \\
\mathsf{switch}(q, q') &= [\exists a \in A : \rho(q)_a \neq \rho(q')_a] \\
\mathsf{exit}(q) &= [\rho(q) \equiv \mathsf{CI}^A] \\
\mathsf{payoff}(q) &= \mathsf{viol}(q) \qquad \text{(by } \texttt{berne_property_classes.py:71}\text{)}
\end{aligned}$$

switch is defined on consecutive-state pairs; when applied to a history it is understood as the disjunction over consecutive-pair occurrences. This resolves the AP typing gap noted in earlier drafts.

The abstraction is deliberately thin. Every component is either externally motivated (state space, alphabet, transitions from the corpus of Section 7) or serves a specific formal role (controllability, payoff, interventions in Section 5).

The identification payoff := viol is a *modelling assumption* (assumption A1), imported from the corpus model. Its consequences are documented in Section 8 and analysed critically in Sections 9 and 16.

5. Observation, intervention, and the antithesis window

The observation function O in Definition 4.1 is deliberately undertyped so that we can now distinguish several standard observer models:

Definition 5.1 (Observer models)**.** An *observer model* is a function $O : \mathrm{Hist}(\mathcal{G}_g) \to \Gamma^*$ over a per-observer observation alphabet Γ. We use:

- $O_{\mathrm{ext}}(h) = (\rho(q_0), \rho(q_1), \ldots, \rho(q_n))$ — the sequence of role configurations. Forgets all events; forgets all annotations.
- $O_{\mathrm{conv}}(h) = (\rho(q_i), \mathrm{ann}^{\mathrm{conv}}(h)_i)_i$ — the external analyst plus the per-tick conversational-pattern label from Definition 4.3. Anti-oracle by construction: the pattern dictionary maps role-configurations (not game identity) to a pattern name.
- $O_a(h) = (\rho(q_i)_a, \mathrm{ann}^{\mathrm{sub}}_a(h)_i)_i$ for $a \in A$ — participant a's per-tick role plus their private substrate annotation. Does not reveal other agents' substrate; does not reveal game identity.
- $O_\infty(h) = h$ plus g's identity — an ORACLE reference point, used only for formal comparison. Never used to define t_{obs} for a live discrimination task.

Definition 5.2 (Anti-oracle admissibility)**.** An observer O is *anti-oracle* for the discrimination task $(g, \mathfrak{A})$ if it is a proper function of h that never leaks g's identity or the payoff status of h. Formally: for every $h, h' \in \mathrm{Hist}$, if h and h' agree on the projection (ρ, ann) that O inspects, then $O(h) = O(h')$. The observers O_{ext}, O_{conv}, and O_a are anti-oracle; O_∞ is (by design) not.

The observer choice is not a stylistic matter: it changes what t_{obs} evaluates to (Section 8).

Definition 5.3 (Observational identifiability)**.** Let $\mathfrak{A} \subseteq \{\text{games in the corpus}\} \setminus \{g\}$ be an *alternative class*. The observational-identifiability frontier of g under $(\mathfrak{A}, O)$ is

$$t_{\mathrm{obs}}(g;\, \mathfrak{A},\, O) \;=\; \min\Big\{\, k \;\Big|\; \forall g' \in \mathfrak{A} :\; O(\tau_{\le k}) \text{ is inconsistent with every admissible } \tau' \text{ of } g' \Big\}$$

with $t_{\mathrm{obs}}(g; \mathfrak{A}, O) = \perp$ (undefined) if no such k exists.

t_{obs} is meaningful only relative to an alternative class. Throughout this paper $\mathfrak{A}$ defaults to the seven other canonical Berne games in the corpus; when a different alternative class is meant we write $t^{\mathrm{role}}_{\mathrm{obs}}$, $t^{\mathrm{model}}_{\mathrm{obs}}$, etc. This is a weaker interpretation than universal game identifiability — it measures *corpus-discriminability* — and we say so throughout.

Definition 5.4 (Intervention class)**.** An *intervention class* $\mathcal{I}$ is a subset of edit-automata primitives $(\Sigma^* \to \Sigma) \cup \{\mathsf{suppress}, \mathsf{insert}\}$ in the sense of Falcone, Fernàndez, and Mounier [FFM12]. The default intervention class $\mathcal{I}_{\text{pol}}$ is one-tick single-agent policy switch:

$$\mathcal{I}_{\text{pol}} = \{\text{apply role reassignment } r \in R \text{ to agent } a \in A \text{ at some } t\}.$$

Definition 5.5 (Intervention sub-types)**.** Given $\mathcal{I} \subseteq (\Sigma^* \to \Sigma)$, we distinguish three sub-classes of single-agent policies:

- $\mathfrak{N}^{\pi} = \{\pi \in \mathcal{I} : \pi \text{ is history-independent and single-agent}\}$ — a fixed action attached to a single agent, applied from time 0 (*pre-committed policy*).
- $\mathfrak{N}^{\Diamond} = \{\pi \in \mathcal{I} : \pi \text{ is conditioned on a recognition event } \varepsilon \in \Sigma_O\}$ — the policy activates only after the observer receives a discriminating signal (*cascading policy*: recognition → activation).
- $\mathfrak{N}^{E} = \{\pi \in \mathcal{I} : \pi \text{ is conditioned on an evidence accumulator } E \text{ crossing a threshold } K\}$ — the policy activates on $E(\tau_{\le t}) \ge K$ (*evidence-block policy*).

These sub-classes are not disjoint: a single-agent history-independent policy is in $\mathfrak{N}^{\pi}$; a policy that activates after both a recognition event AND crosses an evidence threshold lies in $\mathfrak{N}^{\Diamond} \cap \mathfrak{N}^{E}$. The intended interpretation is that each sub-class is the smallest class that captures the mechanism of a particular per-game antithesis (see Section 8).

Definition 5.6 (Monitor machine)**.** Following the Bauer–Leucker–Schallhart runtime-verification and Ligatti–Bauer–Walker edit-automata traditions, an *evidence-side monitor machine* is a tuple $\mathcal{M} = (M, m_0, \mu, \gamma)$ where M is a finite set of *monitor states*, $m_0 \in M$ is the initial state, $\mu : M \times \Sigma \to M$ is the update function on observable events, and $\gamma : M \to \{\bot, \top, ?\}$ is a three-valued verdict map. The state set M is per-game: $M_g = \{*\}$ (singleton, the observer already suffices) for $g \in \{\text{WDYB}, \text{NIGYSOB}, \text{IWFY}, \text{KICK_ME}, \text{SWYMD}\}$; and $M_g = \{0, 1, \ldots, K_g\}$ (counter to threshold K_g) for $g \in \{\text{TBFDP}, \text{SCHLEMIEL}\}$. The evidence accumulator E invoked by $\mathfrak{N}^{E}$ is exactly m under μ; its threshold-crossing predicate is γ. *Terminological convention.* $\mathcal{M}$ denotes the *machine*; $m \in M$ denotes an *element* of M (a monitor state); the per-agent instance at round r is written m_a^r. The observer of Definition 5.1 is the derived map $\gamma \circ \mu^*$; Definition 5.6 makes explicit the machine that produces it.

Definition 5.7 (Per-agent state)**.** The *per-agent state* at round r is $X_a^r := (\rho_a^r, m_a^r) \in \{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}, \mathsf{Cl}\} \times M_g$ where ρ_a^r is the role coordinate (Definition 4.1) and $m_a^r \in M_g$ is the per-agent monitor state of Definition 5.6. The joint carrier $\prod_{a \in A} X_a^r$ refines Q_g of Definition 4.1 in the monitor slot only; when $M_g = \{*\}$ the refinement is trivial and $X_a^r \equiv \rho_a^r$, so Remark 4.2's minimality is preserved on the five singleton-monitor games.

Definition 5.8 (Intervention locus)**.** The *locus map* λ tags each intervention sub-type of Definition 5.5 by two disjoint typed sorts:

$$\lambda_{\text{Trigger}} : \mathcal{I}_{\text{sub}} \to \{\mathsf{precommit}, \mathsf{recognition\text{-}}\varepsilon, \mathsf{evidence\text{-}threshold}\},$$
$$\lambda_{\text{Register}} : \mathcal{I}_{\text{sub}} \to \{\mathsf{trivial}, \mathsf{counter}\},$$

where $\mathcal{I}_{\text{sub}} = \{\mathfrak{N}^{\pi}, \mathfrak{N}^{\Diamond}, \mathfrak{N}^{E}\}$ and per-sub-type population is

	λ_{Trigger}	$\lambda_{\text{Register}}$
$\mathfrak{N}^{\pi}$	precommit	trivial
$\mathfrak{N}^{\Diamond}$	recognition-ε	trivial
$\mathfrak{N}^{E}$	evidence-threshold	counter.

The two sorts are disjoint (*Trigger* is a firing-condition on observable events; *Register* names the monitor-state alphabet supporting the trigger) and both are non-trivial on the corpus (each sort takes ≥ 2 distinct values across the seven games; Table 4). No object is introduced beyond Definitions 5.5 and 5.6.

Definition 5.9 (Admissible continuation)**.** Given a history-prefix $h_{\leq k}$ and a policy $\pi \in \mathcal{I}$ that dictates the agent's next controllable action, the set $\mathrm{Cont}_u(h_{\leq k}, \pi)$ of *admissible uncontrollable continuations* is the set of extensions $h' = h_{\leq k} \cdot \pi(h_{\leq k}) \cdot e_{k+1} \cdot q_{k+1} \cdots$ in which every e_j with $j > k$ is either the controllable move dictated by π at step j or any uncontrollable event $e_j \in E_g^u$ enabled at q_j.

Definition 5.10 (Redirection frontier)**.** The redirection frontier of g under $(\mathcal{I}, P)$ against an adversarial uncontrollable environment is

$$t_{\mathrm{redir}}(g;\, \mathcal{I},\, P) \;=\; \max\Big\{\, k \;\Big|\; \exists \pi \in \mathcal{I} :\; \forall\, h' \in \mathrm{Cont}_u(h_{\leq k}, \pi) :\; h' \notin P \Big\}$$

with $t_{\mathrm{redir}} = \bot$ if no such k exists. The quantification is existential over π (there exists a specific policy) and universal over uncontrollable continuations (guaranteed avoidance). When $E_g^u = \emptyset$ (the case throughout the current corpus), the universal quantifier is over the singleton $\{\pi\text{-extension}\}$ and the definition specialises to "reactive intervention at time k prevents P".

Definition 5.11 (Antithesis window)**.** The (signed) antithesis window of g under $(\mathfrak{A}, O, \mathcal{I}, P)$ is a partial function

$$W : \mathrm{Games} \times \mathrm{AltClass} \times \mathrm{Obs} \times \mathrm{Intv} \times \mathrm{Payoff} \;\rightharpoonup\; \mathbb{Z} \cup \{\pm\infty\},$$

undefined when either $t_{\mathrm{obs}}(g; \mathfrak{A}, O) = \bot$ or $t_{\mathrm{redir}}(g; \mathcal{I}, P) = \bot$, and equal to $t_{\mathrm{redir}}(g; \mathcal{I}, P) - t_{\mathrm{obs}}(g; \mathfrak{A}, O)$ otherwise. The arithmetic on the extended value set is fixed by the following table (writing $\bot$ for the undefined image of W):

W	$t_{\mathrm{redir}}{=}\bot$	$t_{\mathrm{redir}}{\in}\mathbb{Z}$	$t_{\mathrm{redir}}{=}+\infty$	$t_{\mathrm{redir}}{=}-\infty$
$t_{\mathrm{obs}}{=}\bot$	$\bot$	$\bot$	$\bot$	$\bot$
$t_{\mathrm{obs}}{\in}\mathbb{Z}$	$\bot$	$t_{\mathrm{redir}} - t_{\mathrm{obs}}$	$+\infty$	$-\infty$
$t_{\mathrm{obs}}{=}+\infty$	$\bot$	$-\infty$	$\bot$	$-\infty$
$t_{\mathrm{obs}}{=}-\infty$	$\bot$	$+\infty$	$+\infty$	$\bot$

The two $\bot$-diagonal cells $(+\infty, +\infty)$ and $(-\infty, -\infty)$ correspond to indeterminate arithmetic on the affinely-extended integers and are treated as $W = \bot$ (partial-function convention: no image is assigned).

Remark 5.12 (W is not a formal-methods novelty). t_{obs} specialises the diagnosability-delay concept of Sampath et al. [SSL$^+$95] to game identification. t_{redir} specialises the reaction-bound-for-coprognosis concept of Kumar and Takai [KT10]. W is their signed difference; Kumar and Takai's paper explicitly proves that the diagnosability and prognosability axes are incomparable, so the existence of games with $W < 0$ is folklore in DES engineering. We do not claim W as a formal-methods invariant. What we do claim is that W discriminates classical Berne games in a substantive way (Sections 8 and 9) and aligns with Berne's own antithesis-language distinction between reactive and pre-committed interventions.

6. Role projection and switch algebra

Neither Berne 1964 nor Karpman 1968 formulates the following construction; it is a project abstraction over Karpman's role vocabulary. Two per-agent role transitions between consecutive states induce a total map on the role-multiset. When this map is a bijection, it is an element of the symmetric group S_3 over $\{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}\}$.

Definition 6.1 (Induced permutation)**.** For a game g with a role trace $\rho(\tau_g) = r_0 r_1 \cdots r_n$ whose opening r_0 and payoff-locked terminal r_n share the same role multiset (viewed as a multiset over $\{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}\}$), the *induced permutation* $\sigma_g \in S_3$ is the (unique) permutation on the role multiset that maps r_0 to r_n.

The computation is implemented in the accompanying software artefact (Appendix E); machine-checked coverage over the four canonical games with dedicated switch-algebra tests yields:

Proposition 6.2 (S_3 generation, computational witness)**.** *The switch permutations of Why Don't You Yes But and Now I've Got You You Son of a Bitch generate the full symmetric group:* $\langle \sigma_{\mathrm{WDYB}}, \sigma_{\mathrm{NIGYSOB}} \rangle = \langle (\mathsf{Vic\ Per\ Res}), (\mathsf{Per\ Res}) \rangle = S_3$.

Proof sketch. Direct computation: (Per Res) and (Vic Per Res) generate S_3 by inspection of the six permutations. Machine-verified in the artefact of Appendix E. □

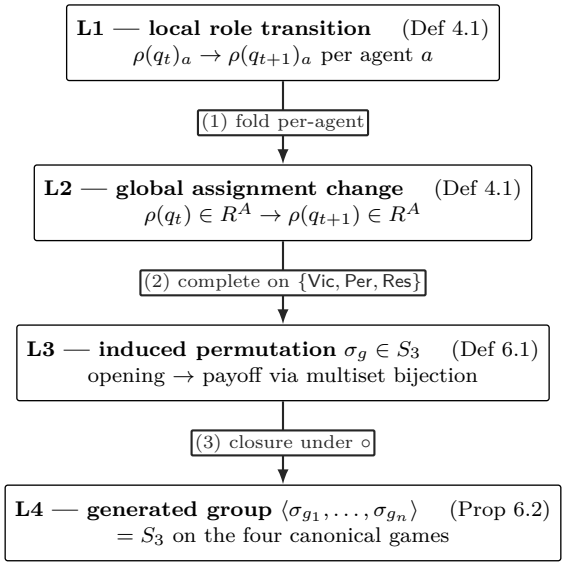


Figure 2. Switch hierarchy. Four numbered construction steps (1)–(3) build the chain L1 → L2 → L3 → L4. Local per-agent transitions (L1) fold into a global assignment change (L2); the global change completes to a permutation $\sigma_g \in S_3$ (L3); the group generated by the switch permutations of the games is S_3 (L4). Each numbered arrow is a construction step defined in the paper; none is a trivial identification. Grayscale-safe: distinctions carried by borders, geometry, and numbered labels only.

7. Corpus formalisation

We formalise six classical Berne games from [Ber64] plus one project-derived variant. Every trace is machine-checked in the accompanying software artefact (see Appendix E).

Game	Source anchor	Level	σ_g	Type
WDYB	[Ber58] + [Ber64]	P0+S1	(Vic Per Res)	classical
IWFY	[Ber64]	S1	(Vic Per Res)	classical
SWYMD	[Ber64]	S1	(Vic Per Res)	classical
NIGYSOB	[Ber64]	S1	(Per Res)	classical
KICK_ME	[Ber64]	S1	(Per Res)	classical
SCHLEMIEL	[Ber58] + [Ber64]	P0+S1	(Per Res)	classical
TBFDP	project-derived variant‡	–	(Vic Per)	project-derived

‡ TBFDP is a project-derived betrayal-detection variant; its game structure is generic transactional [Ber64], the specific betrayal-searcher story is programme-original. Full role traces (opening → payoff-locked) are given in Appendix B.

Table 1. The seven games modelled in this paper. See Appendix B for complete traces including fluent sequences.

Role projection is ours. Every role label in Table 1 is a *project-original projection* of Berne's transactional structure onto Karpman's role space, in the sense of Section 2. Karpman 1968 does not analyse these games.

TBFDP is not canonical Berne. We include TBFDP as a project-derived control because it is the only game in the programme with a receiver-dependent payoff channel (via evidence accumulation); this makes it a useful contrast to the six canonical games whose payoff is receiver-independent under $\mathsf{payoff} := \mathsf{viol}$.

8. Antithesis signatures

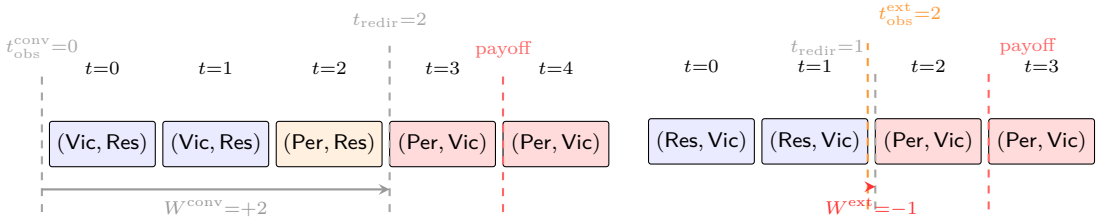
For each game we compute the antithesis window under the default model $(\mathfrak{A}_{\text{dataset}}, O_{\text{ext}}, \mathcal{I}_{\text{pol}}, P_{\text{lock}})$. Table 2 summarises; per-observer values across the four observer profiles are recorded in the accompanying software artefact (Appendix E).

Game	$t_{\text{obs}}^{\text{ext}}$	t_{redir}	W^{ext}	Antithesis class
WDYB	$\perp^\star$	2	$\perp$	UNDEFINED
IWFY	$\perp^\star$	2	$\perp$	UNDEFINED
SWYMD	$\perp^\star$	2	$\perp$	UNDEFINED
NIGYSOB	1	1	0	KNIFE-EDGE
KICK_ME	3	1	-2	PRE-COMMITTED
SCHLEMIEL	3	3	0	KNIFE-EDGE
TBFDP	1	2	$+1$	REACTIVE (evidence-block)

* WDYB/IWFY/SWYMD share an identical role trace: role-level identification is not possible.

TABLE 2. Antithesis windows for the corpus, under the default model $(\mathfrak{A}_{\text{dataset}}, O_{\text{ext}}, \mathcal{I}_{\text{pol}}, P_{\text{lock}})$. Values are computationally reproducible from the accompanying software artefact (Appendix E); every value carries an observer superscript; the classification depends critically on the alt-class choice (see Remark 8.1).

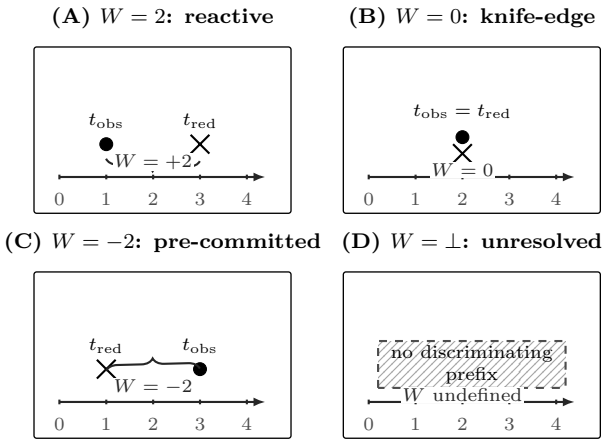
Remark 8.1 (Alt-class dependence of the classification). The values in Table 2 are computed under the strict alternative class $\mathfrak{A}_{\text{dataset}}(g) = \text{Corpus} \setminus \{g\}$ of the six canonical Berne games plus TBFDP. Under this strict class, NIGYSOB has $W = 0$ (its opening $(\mathsf{Res}, \mathsf{Vic})$ is unique in the corpus, so $t_{\text{obs}} = 1$; the same as t_{redir}). Under a wider alternative class $\mathfrak{A}_{\text{nongame}} \supseteq \mathfrak{A}_{\text{dataset}}$ that admits generic helpful transactions with opening $(\mathsf{Res}, \mathsf{Vic})$, NIGYSOB's t_{obs} shifts to 2 (only observable after White pounces) and W becomes -1: the paradigmatic negative-window case of the informal prose reading. **The paper's REACTIVE / PRE-COMMITTED story is therefore alt-class-dependent**, and under strict $\mathfrak{A}_{\text{dataset}}$ it is KICK_ME (not NIGYSOB) that realises the paradigm PRE-COMMITTED regime with $W = -2$. Table 2 reports the strict-$\mathfrak{A}_{\text{dataset}}$ values because these are the values the code can reproduce without further modelling; the wider-alt-class values are deferred to future work that includes a formal $\mathfrak{A}_{\text{nongame}}$.



(A) WDYB: positive window. O_{conv} discriminates at t=0 via the "yes, but…" pattern; Others' antithesis is executable up to t=2; payoff locks at t=3.

(B) NIGYSOB: negative window *under* $\mathfrak{A}_{\text{nongame}}$. Under a wider alternative class that admits generic helpful transactions sharing the opening (Res, Vic), the game is only observable at t=2 (post-pounce), while the last executable reactive intervention would be at t=1. Under strict $\mathfrak{A}_{\text{dataset}}$, NIGYSOB's opening is uniquely discriminating and $W = 0$; see Remark 8.1. KICK_ME (not shown) is the paradigm strict-$\mathfrak{A}_{\text{dataset}}$ negative-window case ($W = -2$).

FIGURE 3. Positive (WDYB) vs negative (NIGYSOB) antithesis windows, under the default $\mathcal{I}_{\text{pol}}$ intervention class. The NIGYSOB arrow reverses direction relative to WDYB: the redirection frontier lies to the LEFT of the observation frontier, so no reactive intervention conditioned on discriminating evidence can prevent the payoff.



filled disc • marks t_{obs} (observation of an $\mathfrak{A}$-discriminating prefix); cross × marks t_{red} (last effective intervention / redirection instant); bra

FIGURE 4. Four regimes of the antithesis window $W = t_{\text{red}} - t_{\text{obs}}$ under a shared horizontal time axis $\{0, 1, 2, 3, 4\}$. **(A)** $W > 0$: reactive intervention feasible after game identification (obs at $t = 1$, red at $t = 3$; brace shows $W = +2$). **(B)** $W = 0$: knife-edge (coincidence at $t = 2$). **(C)** $W < 0$: observation only after the reactive window has closed — only pre-committed policies work (obs at $t = 3$, red at $t = 1$; brace shows $W = -2$). **(D)** $W = \perp$: no observer-under-$\mathfrak{A}$ prefix uniquely discriminates the game (hatched interval). Same time-axis grammar, same event markers, same label placement across all four panels; grayscale-safe.

REACTIVE class (under strict $\mathfrak{A}_{\text{dataset}}$). TBFDP is the only member of the corpus with $W^{\text{ext}} > 0$ under strict corpus-discriminability, because its opening $(\mathsf{Per}, \mathsf{Vic})$ is unique in the corpus. Under a wider alt-class $\mathfrak{A}_{\text{conv}}$ (the pattern-conversational observer O_{conv} applied to the game family — see Remark 8.1), WDYB, IWFY, and SWYMD also join this class because the "yes, but..." pattern discriminates them at $t = 0$. For each of these Berne's antitheses are executable after partial game evidence, e.g. for WDYB the Antithesis paragraph in [Ber64] prescribes not-playing-ITHY in response and returning the problem to White The recognition event ε conditioning $\mathfrak{N}^{\Diamond}$ for these games is the observer's per-tick pattern classification (Definition 5.1, O_{conv}), not a per-agent monitor register: no internal cognitive machinery is modelled beyond what is exposed by the observer, and the abstraction boundary is atomic-recognition.

PRE-COMMITTED class (under strict $\mathfrak{A}_{\text{dataset}}$). KICK_ME is the paradigm case with $W = -2$: its opening $(\mathsf{Vic}, \mathsf{Res})$ is shared with WDYB / IWFY / SWYMD / SCHLEMIEL, and the game is distinguishable from SCHLEMIEL only at $t = 3$, well after the reactive-intervention window has closed at $t = 1$. The receiver's a-priori commitment to non-provocation is here a *programme-derived* formalisation of a KICK_ME countermove: the mechanism is stated in [Ber64] (behaviour becomes increasingly provocative until environmental constraints are exceeded), and the formal precommitment is our reconstruction, not a verbatim Berne antithesis. This is a permanent scope statement, not a source-pending

claim. NIGYSOB shifts to $W = -1$ (and joins this class) under the wider alt-class $\mathfrak{A}_{\text{nongame}}$; under strict $\mathfrak{A}_{\text{dataset}}$ it is instead knife-edge because its opening ($\mathsf{Res}, \mathsf{Vic}$) is unique in the corpus. See Remark 8.1. Berne's Antithesis paragraph in [Ber64] for NIGYSOB prescribes "correct behaviour" — the contractual structure stated in detail at the first opportunity, rules strictly adhered to, mistakes avoided, and unexpected contretemps handled without dispute . The pre-pounce interior of NIGYSOB is atomised inside δ_g: no register in the sense of Definition 5.6 is maintained, and no internal threshold-crossing is exposed. This is a deliberate modelling boundary consistent with the source under-attestation of NIGYSOB's antithesis passage in the current citation cache.

KNIFE-EDGE.. SCHLEMIEL. $W = 0$: game identification and last-effective intervention coincide at $t = 3$. Berne's own precursor of the antithesis is Host's refusal to accept the apology-move: from [Ber58], p. 740 (P0), the antithetical response is stated verbatim as *"You can break things and spill things all you like; but please don't say 'I'm sorry!' "*, matching the verbal description on the same page (*"the one who is 'it' breaks things, spills things, and makes messes... and each time says: 'I'm sorry!' "*). Berne 1958 is therefore a P0 anchor for both the game and its antithesis; the earlier source-request marker for Berne 1964 corroboration is retained as a courtesy but not a blocker. The SCHLEMIEL evidence counter of Definition 5.6 is a programme-modelled forgiveness-count: Berne 1958 gives the antithesis directly (*"please don't say 'I'm sorry!' "*) without a numeric threshold, so both the $\mathfrak{N}^{\pi}$ (pre-commitment) branch and the $\mathfrak{N}^{E}$ (threshold) branch of Table 4 formalise plausible implementations of the same source-attested antithesis.

Game	Monitor state m_a^r	Update μ	Trigger predicate γ
TBFDP	$m^r \in \{0, \dots, K\}$, K=3	$m^{r+1} = m^r + [\text{betray} \wedge \text{sample}]$	$\gamma = \top \iff m \geq K$
SCHLEMIEL	$m^r \in \{0, \dots, K\}$	$m^{r+1} = m^r + [\textsc{apologise}]$	$\gamma = \top \iff m \geq K$

TABLE 3. Register mechanics for the two explicit-register games (Definition 5.6). TBFDP's counter is a code-witnessed per-tick evidence accumulator (independent update rule + two negative controls); SCHLEMIEL's counter is a programme-modelled forgiveness-count with no numeric threshold in the source. Both are single-crossing monitors (no hysteresis needed); no register or trigger machinery is exposed by any other game in the corpus.

Scope of the payoff model on receiver-reaction channels. The payoff model $\mathsf{payoff} := \mathsf{viol}$ of assumption A1 is a *modelling scope*, not a claim about which observable receiver reactions can or cannot influence the payoff of a game under a richer model. Receiver-reaction channels — for example those that couple the observed receiver reaction back into the Persecutor's own payoff state via an enriched Persecutor representation with an internal state that tracks the receiver's response — are a legitimate object of study and are exactly the place where a formal treatment of role-transition mechanisms (role-safety obligations and their admissible relaxations) belongs. The seven-game corpus is compatible with that treatment, but the apparatus needed to develop it (identity-safety obligations, role- transition triggers, the connection to the MCK framework's lock-based semantics) is out of scope for the present paper and is developed as companion work.

9. Cross-game results

The corpus supplies three domain-specific results.

Proposition 9.1 (Role-level under-determination — encoding-relative)**.** *Under the trace encoding of the corpus of Section 7, the role-configuration traces of Why Don't You Yes But, If It Weren't For You, and See What You Made Me Do are pointwise-identical:*

$$\rho(\tau_{\mathrm{WDYB}}) = \rho(\tau_{\mathrm{IWFY}}) = \rho(\tau_{\mathrm{SWYMD}}) = \langle(\mathsf{Vic},\mathsf{Res}),(\mathsf{Vic},\mathsf{Res}),(\mathsf{Per},\mathsf{Res}),(\mathsf{Per},\mathsf{Vic}),(\mathsf{Per},\mathsf{Vic})\rangle.$$

Consequently, for each g in this triple, $t^{\mathrm{ext}}_{\mathrm{obs}}(g;\{\mathrm{WDYB},\mathrm{IWFY},\mathrm{SWYMD}\},O_{\mathrm{ext}}) = \bot$.

Proof. Direct inspection of the corpus of Section 7: WDYB trace at line 182, IWFY at line 276, SWYMD at line 351. All three coincide as multisets in every tick. The role-configuration trace under O_{ext} is precisely the sequence of ρ values, hence $O_{\mathrm{ext}}(\tau)$ coincides for all three games; no prefix distinguishes them. □

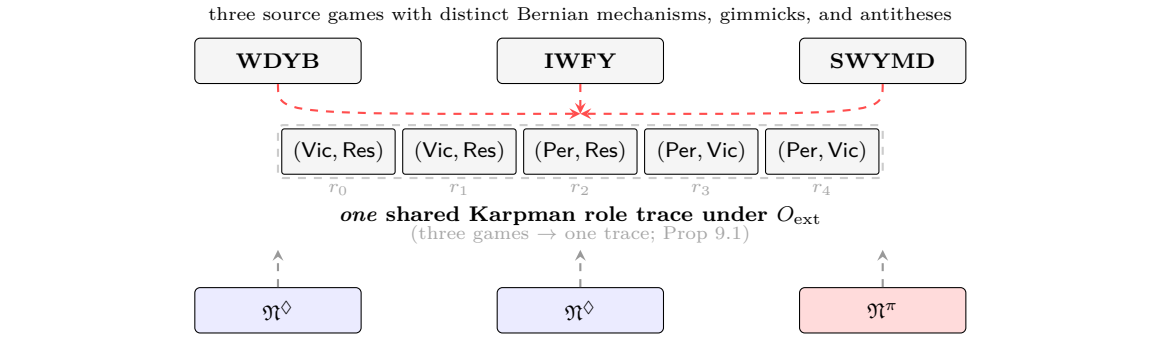


Figure 5. Role information loss (three-into-one). Three distinct Bernian games (top) project to *one* shared Karpman role trace $r_0r_1r_2r_3r_4$ under O_{ext} (centre band, dashed red arrows converging on the same target): the role abstraction is lossy (Prop 9.1). Below the trace, sub-type classification ($\mathfrak{N}^{\Diamond}$ recognition-conditioned, $\mathfrak{N}^{\pi}$ precommit) recovers a distinction between WDYB/IWFY (both $\mathfrak{N}^{\Diamond}$) and SWYMD ($\mathfrak{N}^{\pi}$), showing that intervention structure is not a function of role trace alone (Prop 9.6).

Remark 9.2 (Scope of Proposition 9.1). Proposition 9.1 is a property of the specific programme encoding, not a source-attested claim about Berne 1964. A finer encoding — one that extends the role tuple with, for example, a dwell-time counter or a gimmick-string — could distinguish the three games at role level. The philosophical content — that the Karpman $\{\mathsf{Vic},\mathsf{Per},\mathsf{Res}\}$ alphabet alone is a lossy abstraction of Bernian transactional structure — survives the encoding qualifier because it is exactly the freedom the encoding exercises. Whether Berne 1964's actual game descriptions *force* this identity is an open question; we do not claim it.

Corollary 9.3 (Karpman projection is non-injective on the corpus)**.** *Let $\Pi_K : \mathrm{Hist}(\mathcal{G}) \to \{\mathsf{Vic},\mathsf{Per},\mathsf{Res},\mathsf{CI}\}^{A\times *}$ be the Karpman role projection $\Pi_K(h) = (\rho(q_0),\ldots,\rho(q_n))$ that forgets events and annotations. Then Π_K restricted to the corpus histories is non-injective:*

$\Pi_K(h_{\text{WDYB}}) = \Pi_K(h_{\text{IWFY}}) = \Pi_K(h_{\text{SWYMD}})$ *while* $h_{\text{WDYB}}, h_{\text{IWFY}}, h_{\text{SWYMD}}$ *are distinct executions of distinct games.*

Proof. Immediate from Proposition 9.1: the three histories have identical ρ-projections but differ in their game identity (equivalently in their annotation streams ann^{conv}, which map each game to a distinct conversational-pattern trajectory). Hence Π_K is not injective on the corpus. □

Remark 9.4 (Projection reading under Definition 5.7)*.* Under Definition 5.7 the Karpman projection factors as a role-slot projection $\pi_K : X_a^r \mapsto \rho_a^r$, an information-forgetting map (*not* a quotient in the category- theoretic sense: no equivalence relation is named, no quotient object $X/\!\sim_K$ is constructed, no downstream lemma uses one). So Corollary 9.3 is the statement that distinct X-trajectories collapse to the same ρ-trajectory on the WDYB / IWFY / SWYMD sub-corpus: the annotation and monitor-state content that distinguishes the three games lies in the fibres of π_K. This makes the non-injectivity a property of the projection itself rather than of a specific encoding, addressing part of Remark 9.2.

Definition 9.5 (Minimal intervention requirement)**.** For a game g and an intervention sub-class $T \subseteq \mathcal{I}$, we say T is a *minimal successful class* for g under $(\mathfrak{A}, O, P)$ if: (i) some $\pi \in T$ guarantees $\neg P$ against every admissible uncontrollable continuation from a discriminating observation prefix; (ii) no strict sub-class $T' \subsetneq T$ contains such a π; (iii) the source-described antithesis for g (from [Ber64], Antithesis paragraph of the corresponding game entry;) maps to some $\pi \in T$ under our formalisation. We write $T_{\mathcal{A}}(g) = T$ for the minimal successful class of g under the sub-class family $\{\mathfrak{N}^{\Diamond}, \mathfrak{N}^{\pi}, \mathfrak{N}^{E}\}$.

Proposition 9.1 is a corpus-level statement, not a universal theorem. Its philosophical content is more interesting than its proof: at the granularity of the Karpman role projection alone, the Bernian game structure is under-determined; discrimination requires either conversational content (the "yes, but…" phrase for WDYB; "if it weren't for you" for IWFY; "see what you made me do" for SWYMD) or private substrate content (O_a). This is a formal statement of a familiar observation from clinical Transactional Analysis: V/P/R labels underdetermine the game.

Proposition 9.6 (Permutation-only classification refuted)**.** *The minimal successful class* $T_{\mathcal{A}}$ *of Definition 9.5 is not a function of the induced permutation* σ_g *alone. Concretely,* $\sigma_{\text{WDYB}} = \sigma_{\text{SWYMD}} = (\mathsf{Vic\ Per\ Res})$ *yet* $T_{\mathcal{A}}(\text{WDYB}) = \mathfrak{N}^{\Diamond}$ *while* $T_{\mathcal{A}}(\text{SWYMD}) = \mathfrak{N}^{\pi}$*: WDYB requires the receiver to recognise the game via a mid-interaction pattern-match before revising response policy (a recognition-conditioned intervention in* $\mathfrak{N}^{\Diamond}$ *per Definition 5.5); SWYMD admits a purely policy-invariant single-agent commitment held from* $t = 0$ *(an intervention in* $\mathfrak{N}^{\pi}$ *per Definition 5.5 that requires no recognition event).*

Proof. By exhibition. WDYB's antithesis ([Ber64], Antithesis paragraph for WDYB; content-verified summary: not to play a complementary "I'm Only Trying to Help You" and to return the problem to the initiator;) requires the Others-side receiver to recognise the game via a mid-interaction pattern-match, then revise their response policy — a policy in $\mathfrak{N}^{\Diamond}$ per Definition 5.5, conditioned on a recognition event ε = "yes-but pattern observed twice". SWYMD's antithesis ([Ber64], Antithesis paragraph for SWYMD; content-verified summary: First-Degree = leave the player alone; Second-Degree = throw the decision back on White; Third-Degree = professional handling;) is a history-independent single-agent commitment held from $t = 0$ — a policy in $\mathfrak{N}^{\pi}$ per Definition 5.5, with no recognition

dependency. Since $\mathfrak{N}^{\pi} \setminus \mathfrak{N}^{\Diamond} \neq \emptyset$ and both games realise the same σ_g, the sub-type is not a function of σ_g alone. □

Proposition 9.6 shows that a naïve classification "game $\mapsto$ intervention sub-type" mediated by σ_g alone would collapse WDYB and SWYMD into the same class. Sub-type is instead a function of the triple (permutation, which-agent-reframes, trigger). We formalise this at Table 4.

Remark 9.7 (Locus-tag refinement). Under Definition 5.8, the composite tag $(\lambda_{\text{Trigger}}, \lambda_{\text{Register}})$ likewise fails to be a function of σ_g: on the corpus λ_{Trigger} realises three distinct values (precommit, recognition-ε, evidence-threshold) and $\lambda_{\text{Register}}$ realises two (trivial, counter); each sort is populated by games sharing σ_g. The sub-type $T_{\mathcal{A}}(g)$ (Definition 9.5) determines the composite tag by Definition 5.8, so Proposition 9.6 lifts from a counterexample about a set-valued sub-type to a classification-refinement statement about the composite pair $(T_{\mathcal{A}}(g),\ \lambda(T_{\mathcal{A}}(g)))$.

Remark 9.8 (Capability-profile reading). Under the operational-capability reading of Section 15.5, the WDYB / SWYMD pair of Proposition 9.6 witnesses the same non-injectivity in a capability vocabulary: WDYB's antithesis is driven by observational DIST under O_{conv} (recognition-conditioned $\mathfrak{N}^{\Diamond}$), whereas SWYMD's antithesis uses only CHOOSE and INFL from $t = 0$ (precommit $\mathfrak{N}^{\pi}$) with no DIST-driver. The two games share $\sigma_g =$ (Vic Per Res) yet differ in their capability profile; this is an interpretive restatement of Proposition 9.6, not a new theorem.

Game	σ_g	Sub-type	λ_{Trigger}	$\lambda_{\text{Register}}$
WDYB	(Vic Per Res)	$\mathfrak{N}^{\Diamond}$	recognition-ε	trivial
IWFY	(Vic Per Res)	$\mathfrak{N}^{\Diamond}$	recognition-ε	trivial
SWYMD	(Vic Per Res)	$\mathfrak{N}^{\pi}$	precommit	trivial
NIGYSOB	(Per Res)	$\mathfrak{N}^{\pi}$	precommit	trivial
KICK_ME	(Per Res)	$\mathfrak{N}^{\pi}$	precommit	trivial
SCHLEMIEL	(Per Res)	$\mathfrak{N}^{\pi} \cup \mathfrak{N}^{E}$	precommit $\vee$ evidence-threshold	counter
TBFDP	(Vic Per)	$\mathfrak{N}^{E}$	evidence-threshold	counter

TABLE 4. Intervention sub-types across the corpus, with the locus tag λ of Definition 5.8 expanded into its two disjoint sorts $(\lambda_{\text{Trigger}}, \lambda_{\text{Register}})$. Each sort takes ≥ 2 distinct corpus values. Remark 9.7 extends Proposition 9.6 to the composite (sub-type, λ).

The three sub-types $\{\mathfrak{N}^{\Diamond}, \mathfrak{N}^{\pi}, \mathfrak{N}^{E}\}$ suffice to classify every game in the corpus, with SCHLEMIEL admitting two concurrently; this exhausts the corpus, not a mathematically defined universe of two-agent transactional games (see Section 16).

10. Case studies as protocol sequences

The corpus results of Section 9 are proved on trace-level inspections; the mechanism behind each result is easier to see when the same games are presented in the message-passing format familiar from secure-computation and protocol-analysis papers. This section gives one presentation per game: the six classical Berne games plus TBFDP. Across all seven figures,

lanes carry local state $s_a^r = X_a^r$ per Definition 5.7 and role labels ρ_a^r; horizontal arrows are inter-agent controllable events in E_g^c; the rightmost lane records the observer projection and its verdict; a bracket beneath the timeline marks the intervention window $[t_{\text{obs}}, t_{\text{redir}}]$ where a reactive antithesis in $\mathcal{I}$ can still redirect the trajectory. The recognition-triggered family (WDYB, IWFY, SWYMD) shares an identical role trace but distinct interaction mechanisms (Proposition 9.1); the precommitment family (NIGYSOB, KICK_ME) exhibits negative- and knife-edge windows; the explicit-register games (SCHLEMIEL, TBFDP) carry a monitor state $m^r \in M_g$ (Definition 5.6). Every symbol carries the same meaning as in Section 4–Section 6.

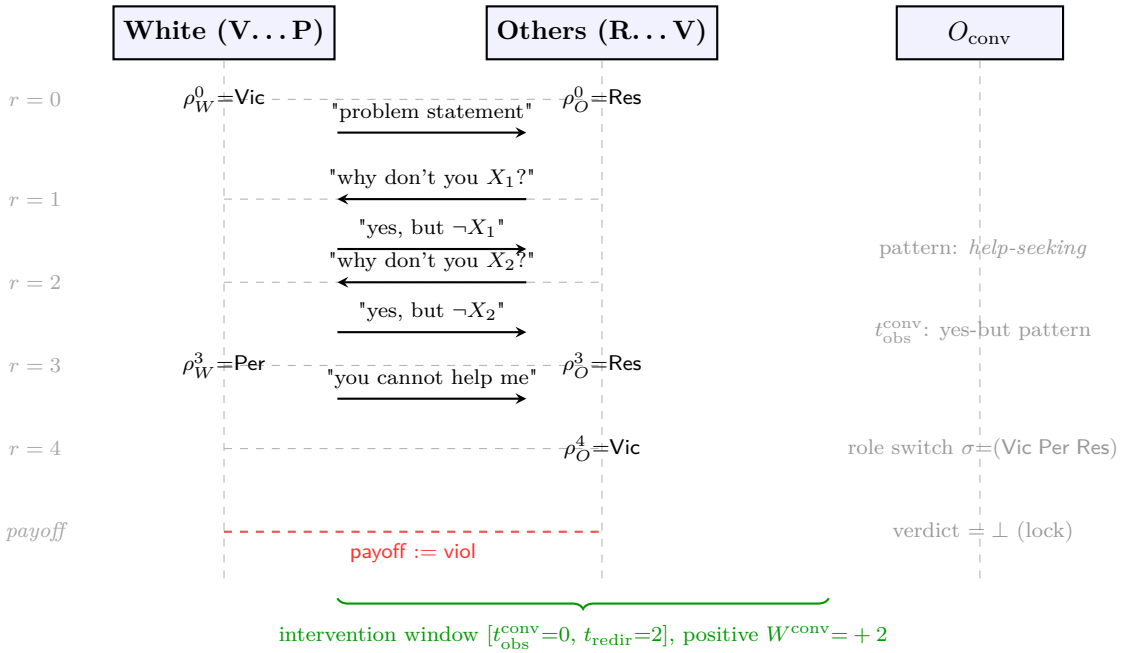


FIGURE 6. WDYB as a message-sequence protocol under O_{conv}. Yes-but pattern is discriminating at $t^{\text{conv}}_{\text{obs}}$=0; Others' antithesis (Appendix A) executable up to t_{redir}=2 before the role switch at r=3. Green bracket: positive window $W^{\text{conv}} = +2$. Switch (Vic Per Res) realised as White's Vic→Per at r=2 composed with Others' Res→Vic at r=3 (Definition 6.1).

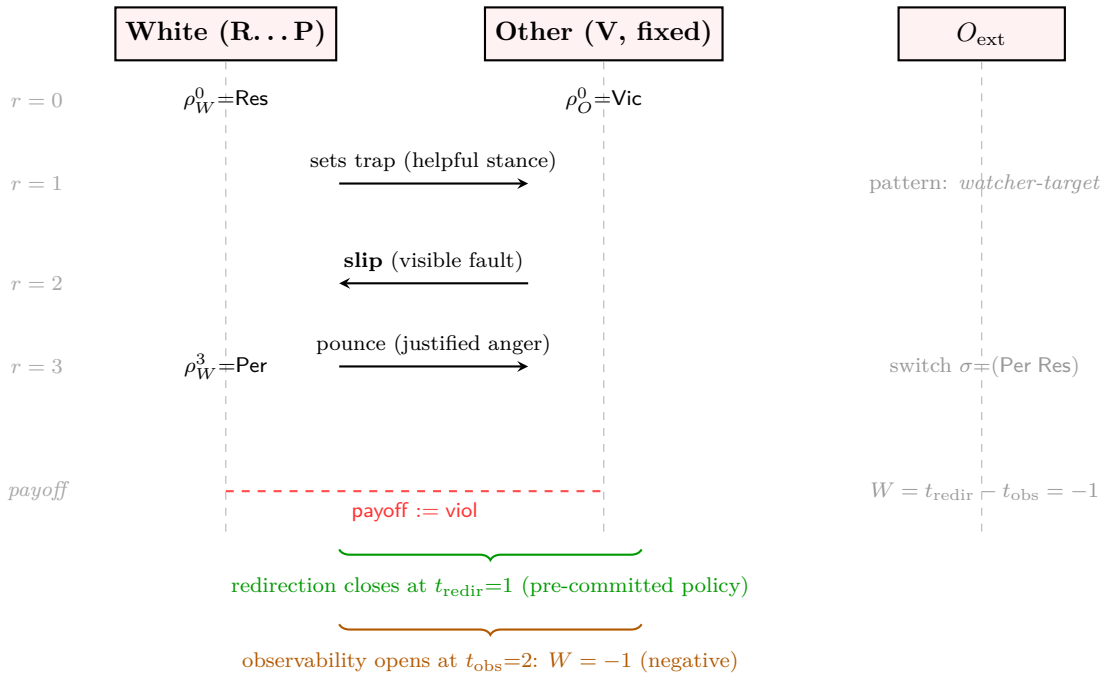


FIGURE 7. NIGYSOB as a message-sequence protocol under $O_{\text{ext}} + \mathfrak{A}_{\text{nongame}}$. Redirection closes at t_{redir}=1; observability opens only at t_{obs}=2: the game is distinguishable only *after* the reactive window has passed (W=−1, Definition 5.11). Berne's antithesis (strict honesty, Appendix A) must therefore be pre-committed, not reactive. Under strict $\mathfrak{A}_{\text{dataset}}$ instead, t_{obs}=1 and W=0 (knife-edge; Remark 8.1).

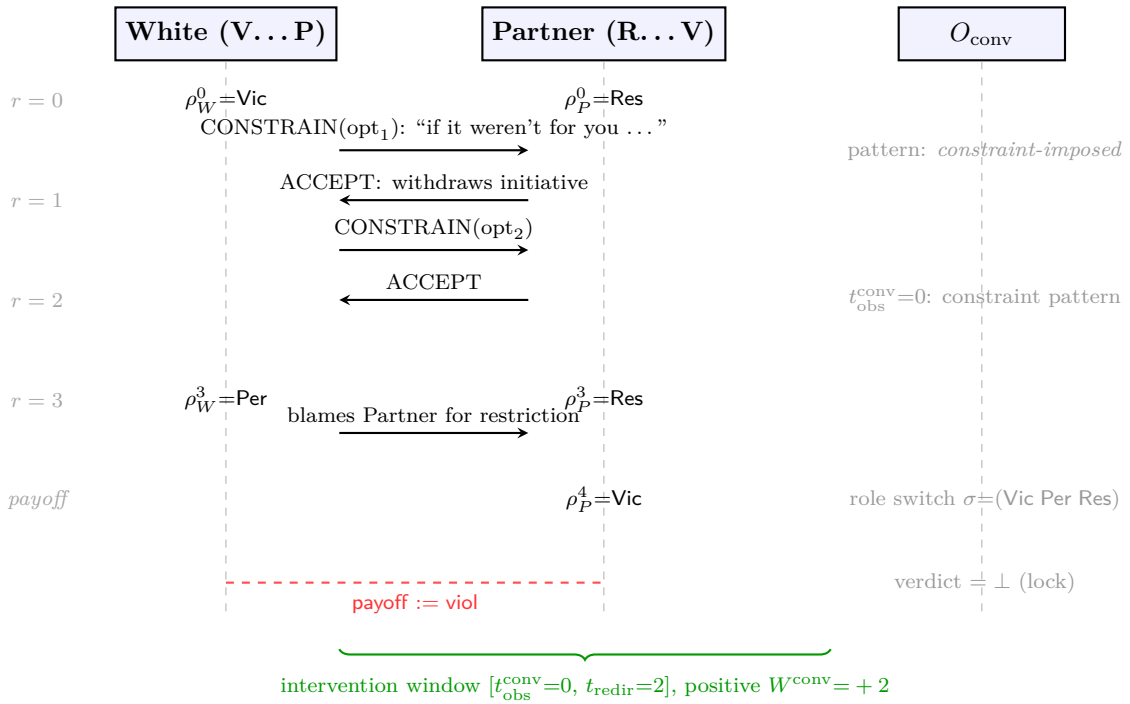


FIGURE 8. IWFY as a message-sequence protocol under O_{conv}. Same π_K-projection as WDYB (Figure 6): $\pi_K(h_{\text{IWFY}}) = \pi_K(h_{\text{WDYB}})$ (Proposition 9.1) — yet the message protocol differs (a CONSTRAIN/ACCEPT restriction/compliance cycle here vs the yes-but suggestion/objection cycle of Figure 6). The recognition-conditioned antithesis (Definition 5.5 $\mathfrak{N}^{\Diamond}$) fires up to t_{redir}=2 before the resentment-payoff role switch at r=3.

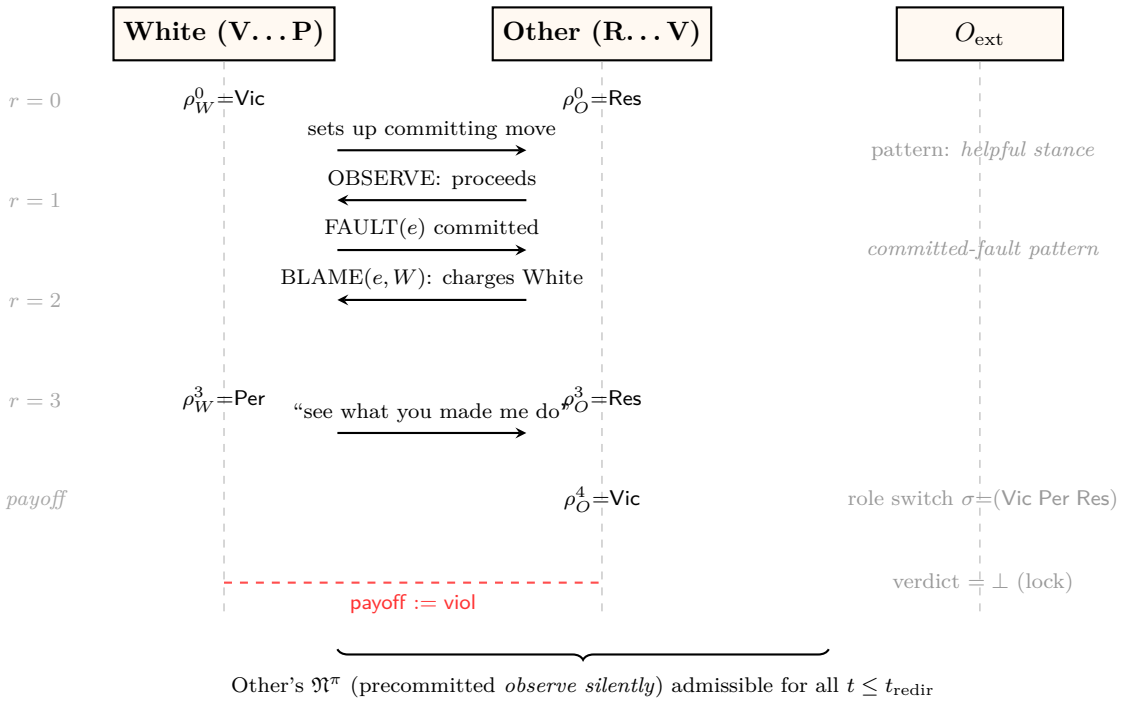


FIGURE 9. SWYMD as a message-sequence protocol under $(O_{\text{ext}}, \mathfrak{A}_{\text{dataset}}, \mathcal{I}_{\text{pol}}, P_{\text{lock}})$; the numeric window entry in Table 2 is $(t_{\text{obs}}^{\text{ext}}, t_{\text{redir}}, W^{\text{ext}}) = (\bot, 2, \bot)$ (UNDEFINED under strict corpus discriminability, since WDYB/IWFY/SWYMD share a role trace under O_{ext}; Proposition 9.1). Same σ_g = (Vic Per Res) as WDYB/IWFY yet the effective antithesis is policy-invariant (Definition 5.5 $\mathfrak{N}^\pi$) rather than recognition-conditioned: Other precommits to non-participation before the fault-blame pattern crystallises. This is the witness for Proposition 9.6: same permutation, distinct sub-type.

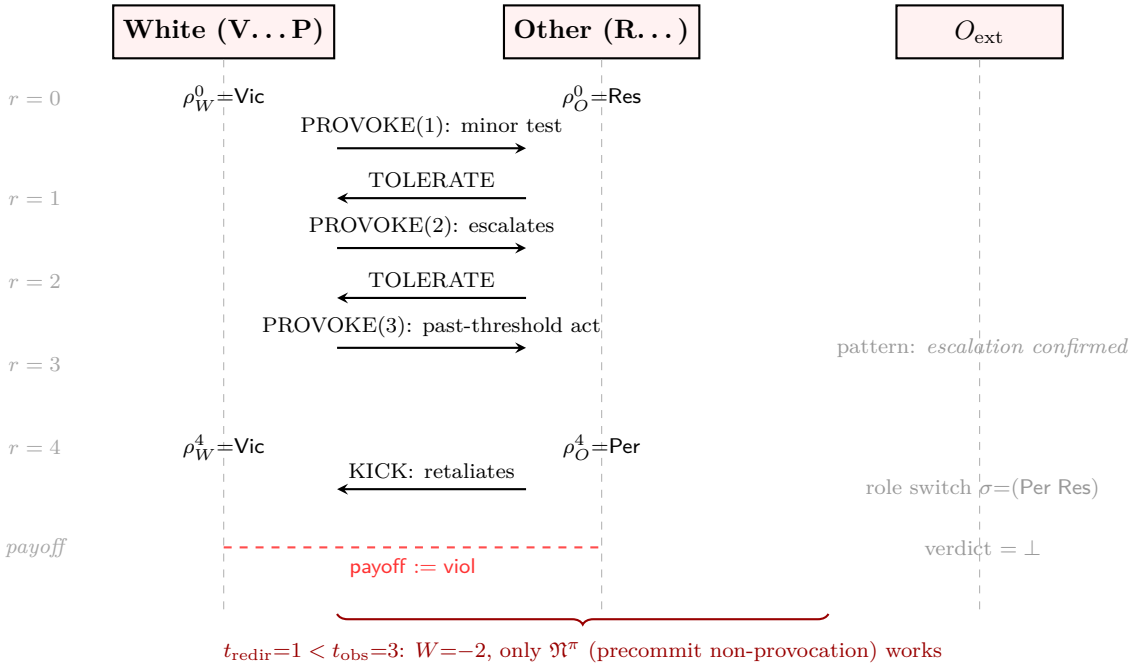


FIGURE 10. KICK_ME as a message-sequence protocol. White's escalation is invisible to O_{ext} until t_{obs}=3; the last effective single-agent redirection (Other precommits to non-provocation) sits at t_{redir}=1. Hence $W = -2$: the paradigm strict-corpus negative-window case. No reactive antithesis is admissible after r=1.

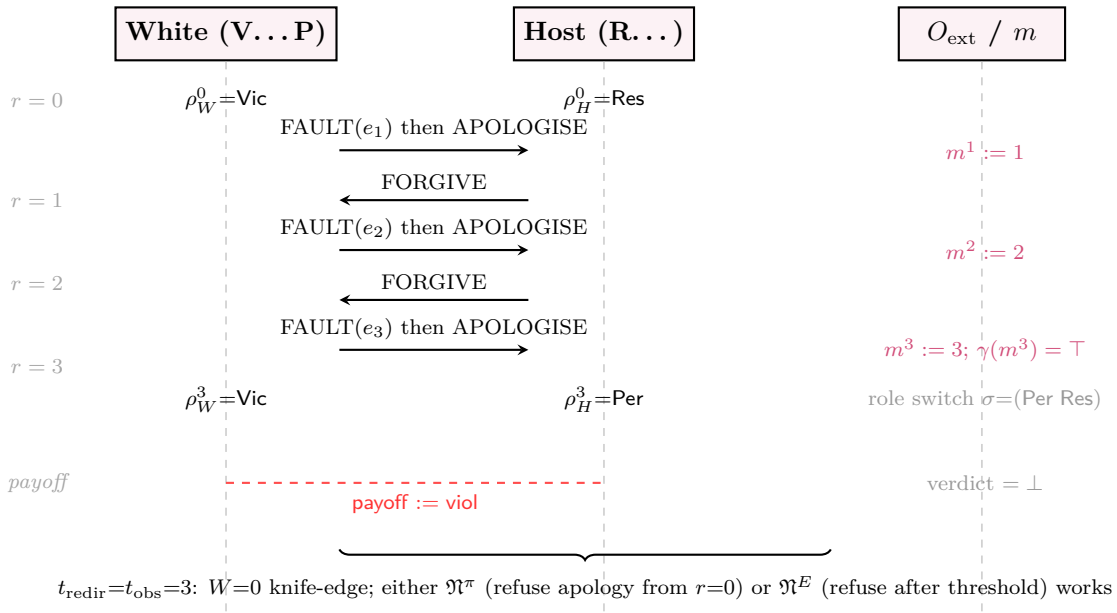


FIGURE 11. SCHLEMIEL as a message-sequence protocol. The Host-side monitor register $m^r \in \{0, \ldots, K\}$ counts accepted apologies (Definition 5.6); the Berne 1958 antithesis (verbatim, Appendix A) is Host's refusal of the apology-move, which formalises as either $\mathfrak{N}^\pi$ (precommit at r=0) or $\mathfrak{N}^E$ (threshold-triggered at $\gamma(m^r) = \top$). The counter is programme-modelled; Berne 1958 gives no numeric threshold.

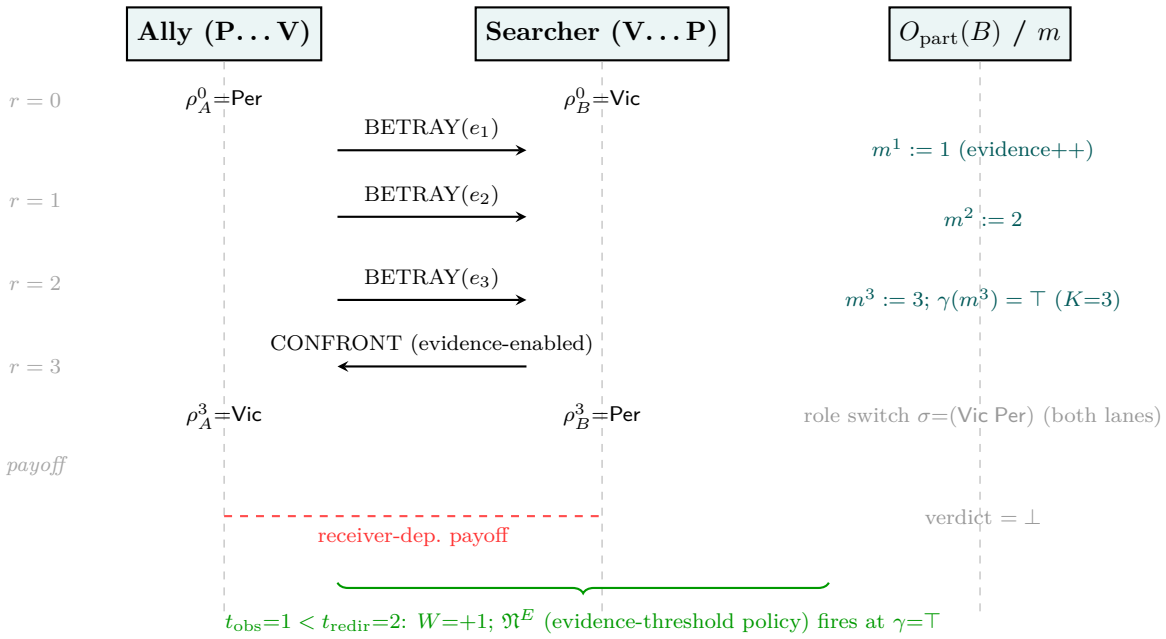


FIGURE 12. TBFDP (*project-derived control*, not a canonical Berne game) as a message-sequence protocol under $O_{\text{part}}(B)$. Searcher's monitor register m^r counts witnessed betrayals under an evidence-accumulation dynamic (Definition 5.6); the threshold-crossing $\gamma(m^r) = \top$ enables $\mathfrak{N}^E$ (Definition 5.5). Unlike the six canonical games, TBFDP has a genuine per-tick evidence update rule and two negative controls in the accompanying artefact (Appendix E).

10.1. **What the sequence view adds.** The message-passing view does three things the corpus tables of Section 8 cannot:

(1) *Locates the intervention window inside a specific round.* In WDYB (Figure 6) the reactive antithesis *fires while the pattern is still forming* — the green band covers rounds $r = 0$–2, and the payoff-locking role switch at $r = 3$ is the round the antithesis must precede. NIGYSOB (Figure 7) shows the opposite geometry: the green band is over before the discriminating orange band even begins. The visual asymmetry is the whole content of the sign-of-W result.

(2) *Ties the two-agent σ-computation to the per-agent timeline.* The 3-cycle (Vic Per Res) in Figure 6 is realised as two consecutive single-agent transitions: White's Vic→Per at r=2 composed with Others' Res→Vic at r=3. The group-theoretic collapse of these into one element of S_3 (Proposition 6.2) hides the temporal separation that the sequence diagram restores.

11. Computational validation

The corpus formalisation is machine-checked against the accompanying software artefact (Appendix E), which verifies well-formedness of every corpus game, the group generation $\langle(\mathsf{Vic}\ \mathsf{Per}\ \mathsf{Res}), (\mathsf{Per}\ \mathsf{Res})\rangle = S_3$ (exhaustive over $|S_3| = 6$), fluent classification for WDYB, IWFY, NIGYSOB, KICK_ME, Hansel–Newton finite-confirmation certificates for the parametrised games, and TBFDP evidence-derived switch tests. Two games (SWYMD, SCHLEMIEL) are absent from the parametrised grids of the auxiliary property-class and Hansel–Newton modules; extending them is a post-drafting task, and the gap does not affect the trace-level results of Section 9 (Section 16). Following the discipline of [Bra26a]: *proof*, *finite witness*, and *unit test* are distinct epistemic registers — Section 9 carries proof status; the machine-check carries computational-witness status over the tested games; the S_3 generation is exhaustive over $|S_3| = 6$.

12. Architecture-side information loss

The previous sections establish that a coarse projection π_K can collapse source-distinct games at the observed-trace level. This section identifies a separate axis on which distinct *internal* agent architectures can be shown to produce the same external role trace, using a model-aware policy discriminator. The material is a compressed absorption of the companion manuscript prepared alongside this paper and is included here in the consolidated version.

Definition 12.1 (Agent architecture)**.** For agent a acting in the interaction system $\mathcal{G}_g$, an *agent architecture* is a tuple $A_a = (X_a^c, M_a^{\mathrm{opp}}, \pi_a)$, where X_a^c is a's local control state, M_a^{opp} is an *opponent-model carrier* (typed entries indexed by peer agents $b \neq a$, values in $\{\bot, *\}$ or an extension thereof), and $\pi_a : X_a^c \times M_a^{\mathrm{opp}} \to E_a^c$ is a deterministic policy.

Definition 12.2 (Model-awareness)**.** Given an agent a and a peer b, A_a is *model-aware* of b if (i) some reachable state carries a defined value $m[b] \neq \bot$ in M_a^{opp}, and (ii) there exist memory maps $m, m' \in M_a^{\mathrm{opp}}$ with $m[b] \neq m'[b]$ and $\pi_a(x, m) \neq \pi_a(x, m')$ for some reachable state x — the policy is sensitive to that entry.

Theorem 12.3 (Same role trace, different architecture)**.** *There exist agent architectures A_0, A_1 and a common schedule $\mathcal{E}$ such that $\pi_K \circ \mathrm{Exec}(A_0, \mathcal{E}) = \pi_K \circ \mathrm{Exec}(A_1, \mathcal{E})$ while A_0 is not model-aware of a peer agent a_2 and A_1 is. Consequently the Karpman role trace is not injective on architecture even at fixed schedule.*

Proof sketch. Fix the two-agent Pounce game with $X_{a_1}^c = \{\mathsf{idle}_1, \mathsf{idle}_2, \mathsf{pounced}\}$ and $M_{a_1}^{\mathrm{opp}}[a_2] \in \{\bot, *\}$. Let A_0's policy ignore $M_{a_1}^{\mathrm{opp}}$ and let A_1's policy read it. Under the schedule $\mathcal{E} = (\mathsf{ok})^4$ both architectures produce the Karpman role trace $\langle \mathsf{Res}, \mathsf{Per}, \mathsf{Per}, \mathsf{Per}\rangle$: A_1 pounces because $m[a_2] = *$ after round 0, A_0 pounces because it always pounces at idle_2. Under the auxiliary schedule $\mathcal{E}' = (\mathsf{fail}, \mathsf{ok}, \mathsf{ok}, \mathsf{ok})$, A_1 visits both $m[a_2] = \bot$ (round 1) and $m[a_2] = *$ (round 2) at idle_2, and $\pi_{a_1}^1$ diverges on that branching pair: wait vs pounce. Model-awareness clause (ii) of Definition 12.2 is witnessed on-trajectory over the extended reachable set, not intensionally. The construction is a standard bisimulation-distinguishing pair [Par81, Mil89]; the added content is Definition 12.2 as a Berne-instantiated policy-input-sensitivity predicate. □

Corollary 12.4 (Corpus-level architecture-side observations)**.** *Across the seven-game corpus, (a) there exist finite witnesses of model-awareness variation at fixed Karpman trace; (b) architectures sharing a Karpman trace need not share an observability class of Definition 5.1; (c) the assignment* role$(a) =$ Vic *does not entail that a lacks the intervention-capability sub-signature of any peer. The witnesses have the same shape as Theorem 12.3: memory-branching over an extended schedule family, followed by a policy sensitivity check.*

Remark 12.5 (Contribution scope). The novel content on this axis is Definition 12.2 as a Berne-instantiated policy-input-sensitivity predicate and its non-implication from the Karpman projection π_K (Theorem 12.3). The witness construction is standard bisimulation folklore [Par81, Mil89]; the corpus embedding is the added value.

13. Three-layer architecture: attribution, recognition, realisation

The consolidated paper's formal core is temporal recognition of projection-induced information loss (Section 9, Section 12). Two adjacent frameworks in the same research programme intersect the paper's scope without formal dependency: *critical-property attribution* [Bra26b] on the observer side, and a *typed agent-action framework* [Bra26e] on the intervention side. This section states each intersection precisely, gives one finite application-labelled attribution witness, and identifies a compact three-action quotient for the action layer. Everything below is labelled either conservative interface or application; the theorem core of this paper does not depend on either framework.

Attribution interface (conservative interface to Critical Properties). The Karpman role trace and the intervention-subtype question presuppose that the observer can attribute a role-relevant property to a specific participant at a specific time on the basis of the observed trace. Following [Bra26b], write $\mathrm{Attr}_a(r, p, t)$ for the attributed status by observer a of property p applied to referent r at time t; the value domain includes a distinguished UNTYP entry for the case in which the referent r is not typed under a's architecture. The attribution fails *before* temporal reasoning proceeds if (i) TypeReq(p) is not met by a's architecture (referent typing), (ii) InfoReq(p, t) is not met (the trace prefix does not carry the discriminating evidence), or (iii) the evaluator for p is unavailable. This decomposition is standard in the awareness / algorithmic-knowledge / functional- observability lineage [FH88, HMV94, FTJ10]; the paper uses it as an interface, not as a novel logic.

Support signature. For every property attribution used in the intervention-subtype analysis of the corpus, the minimal support required from the observer's architecture is $(\mathrm{REF}, \mathrm{OTHERREF}, \mathrm{OBS})$: the observer must have a typed reference to itself, a typed reference to each game participant it distinguishes, and access to the observation channel. The observer does *not* require a FUNC-level self-modelling capability or an ATT-level attitude discrimination beyond the role alphabet $\{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}\}$. In particular, the abandoned broader cross-substrate agentic axis is *not* imported.

Attribution-support witness (APPLICATION). Fix the corpus pair (WDYB, SWYMD) shared by Proposition 9.6 as sharing π_K but requiring distinct intervention subtypes. Two observers can face the same external trace but form distinct intervention-relevant refined states: observer a whose architecture satisfies OTHERREF on both participants (typed reference to Per and to Res as distinct objects) can attribute a role-alternation event to a specific participant and thereby record the recognition-conditioned trigger required by $N^{\Diamond}$ (Definition 5.5); an observer a' whose architecture treats participants as interchangeable tokens (a reduced OTHERREF) records only a role-configuration change and can only fire the precommit-typed N^{π}. The two observers face the same trace but form distinct intervention-relevant refined states. The witness is labelled APPLICATION of standard awareness / functional-observability discipline [FH88, FTJ10], not a new architectural result. The Critical Properties link is a CONSERVATIVE INTERFACE because no theorem of the present paper depends on the (TypeReq, InfoReq, SelfCap) apparatus of [Bra26b]; the paper uses only the reader's ability to attribute properties, which every observer model in the DES/RV literature already provides.

Realisation interface (OPERATIONAL REALISATION via Five Actions). The intervention triggered by the observer is externalised through one or more agent actions in the sense of [Bra26e]: five canonical footprint classes READ, THINK, SPEAK, WRITE, ACT. For the intervention-triggering step in the Berne corpus, the externalisation is a SPEAK event (a conversational turn); the antithesis discipline of Berne treats the intervention move as a linguistic act. The paper does not depend on the full five-action alphabet; a three-action quotient suffices.

Definition 13.1 (Three-action quotient)**.** The three-action quotient $q_{5\to3} : \{\text{READ}, \text{THINK}, \text{SPEAK}, \text{WRITE}, \{\text{READ}, \text{THINK}, \text{RESPOND}\}$ collapses the three externalisation classes SPEAK, WRITE, ACT into a single class RESPOND, keeping READ and THINK unchanged. The quotient realises the standard sense–plan–act decomposition of BDI and reactive-agent architectures [RG95, Woo09], and is not claimed as novel.

Proposition 13.2 (Quotient conservativity for the Berne corpus, observation-level)**.** *For every property, verdict, and intervention subtype used in the corpus results, the observable alphabet does not distinguish among* SPEAK*,* WRITE*, and* ACT*: no property in this paper's family depends on carrier persistence, delayed re-reading, or reflexive re-entry. The three-action quotient $q_{5\to3}$ is therefore observation-equivalence- preserving over the property family used here.*

Proof of Proposition 13.2. The observation follows immediately from the alphabet projection: every property in the corpus results is defined over the projection π_K of the interaction trace to the role-configuration alphabet; no property in this family inspects the specific externalisation carrier. Hence $q_{5\to3}$ is a semantics-preserving alphabet projection on this property family. The proposition is a definitional consequence of the corpus scope and is labelled at observation register rather than as a formal theorem, per its trivial-inspection character. □

Remark 13.3 (Negative control: FA-REENTRY). The quotient $q_{5\to3}$ is *not* semantics-preserving in general. A property class depending on carrier persistence – for example, reflexive re-entry, where a SPEAK event is transient (consumed on receipt) and a WRITE event persists in a shared carrier – separates SPEAK from WRITE at the verdict layer

[Bra26e]. This is the boundary condition of the simplification: the quotient applies only to property families whose observable alphabet is the role-configuration projection, not to properties that inspect externalisation-carrier lifetime. The full five-action alphabet must be retained when the property inspects carriers.

Dependency classification. The cross-framework links are classified as follows.

Framework / work	Role relative to this paper	What it contributes
Critical Properties ([Bra26b])	attribution interface (CONSERVATIVE INTERFACE)	what state predicates p the observing agent can represent and attribute; Section 13 exhibits one application witness.
Intersection algebra of S/C/L/K ([Bra26a])	temporal-property typing layer (FORMAL MACHINERY)	what temporal kind of property Φ_p is: the four-bit signature $\chi(\Phi) = (1[\Phi \in \mathsf{S}], 1[\Phi \in \mathsf{C}], 1[\Phi \in \mathsf{L}], 1[\Phi \in \mathsf{K}])$ places Φ in one of 11 nonempty exclusive cells and fixes the minimum verdict-alphabet rung ($n \in \{1, 2, 3, 4, \infty\}$) at which its verdicts are resolvable.
Observation-Relative Implication ([Bra26d])	observation-model bridge (METHODOLOGICAL PRECEDENT)	given an observation model O, what verdict about Φ is available: monitor verdicts depend jointly on the formula and the observation model. The observer classes $O_{\text{ext}}/O_{\text{conv}}/O_a/O_\infty$ specialise this stance.
Five Actions ([Bra26e])	operational realisation (OPERATIONAL REALISATION)	how a recognised distinction can be externalised as response; the three-action quotient $q_{5\to3}$ is the coarse operational vocabulary this paper uses, with the FAREENTRY carrier-sensitivity result as its boundary.

The four layers are orthogonal: Critical Properties types *state predicates*; `ltl_classes` types *trace properties* derived from those predicates; Implicazione types *monitor verdicts* available for a chosen (Φ, O) pair; Five Actions types the *externalisation carrier* of the response. The S/C/L/K signature is used locally in this paper via Lemma 3.1 and, catalogued in Table 5, indexes the temporal families monitored in the corpus and companion frameworks.

TABLE 5. Temporal-property catalogue across the four programme layers. State predicate p (Critical Properties) is lifted to a trace property Φ_p; the four-bit $\mathsf{S}/\mathsf{C}/\mathsf{L}/\mathsf{K}$ signature $\chi(\Phi_p) = (1[\Phi \in \mathsf{S}], 1[\Phi \in \mathsf{C}], 1[\Phi \in \mathsf{L}], 1[\Phi \in \mathsf{K}])$ places Φ_p in one of the 11 nonempty cells of [Bra26a] §8's totality theorem and fixes the minimum verdict-alphabet rung n at which its verdicts are resolvable. NEW INTEGRATION / SYSTEMATISATION; no novelty for the underlying algebra.

Property (name / owner)	State pred. p	Temporal lift Φ_p	S	C	L	K	rung n	cell
Integrity preservation (CP)	Int_a	$\mathbf{G}\ \mathrm{Int}_a$	1	0	0	1	3	$\mathsf{S} \cap \mathsf{K}$
Certificate acquisition (CP × FA)	Cert_a	$\mathbf{F}\ \mathrm{Cert}_a$	0	1	1	0	3	$\mathsf{L} \cap \mathsf{C}$
Antithesis executed within W (LTL Tri.)	Anti_a	$\mathbf{F}_{\leq W}\ \mathrm{Anti}_a$	1	1	0	0	2	$\mathsf{S} \cap \mathsf{C}$ (clopen)
Repeated recognition (LTL Tri.)	Rec	$\mathbf{G}\,\mathbf{F}$ Rec	0	0	1	1	4	$\mathsf{L} \cap \mathsf{K}$ (recurrence)
Eventual persistence (CP)	Cap_a	$\mathbf{F}\,\mathbf{G}\ \mathrm{Cap}_a$	0	0	1	1	4	$\mathsf{L} \cap \mathsf{K}$ (persistence)
Response-after-recognition (LTL Tri. × FA)	$\mathrm{Rec} \to \mathrm{Resp}$	$\mathbf{G}(\mathrm{Rec} \to \mathbf{F}\ \mathrm{Resp})$	0	0	1	1	4	$\mathsf{L} \cap \mathsf{K}$
Never-externalise-without-condition (FA)	$\mathrm{Ext} \wedge \neg\mathrm{Cond}$	$\mathbf{G}\,\neg(\mathrm{Ext} \wedge \neg\mathrm{Cond})$	1	0	0	1	3	$\mathsf{S} \cap \mathsf{K}$
FA-REENTRY (carrier-inspecting, FA)	$\mathrm{Reentry}_C$	$\mathbf{F}(\mathrm{Speak}_C \wedge \mathbf{F}\ \mathrm{Read}_C)$	0	1	1	0	3	$\mathsf{L} \cap \mathsf{C}$ (per carrier C; the carrier-lifetime clause $\Delta \leq L_C$ from [Bra26e] is a stronger monitorability constraint not encoded in the LTL formula alone)

Coverage of the four families is attested by natural witnesses (in the indicator-vector semantics of χ): S (Integrity preservation, Never-externalise-without-condition, both $\in \mathsf{S} \cap \mathsf{K}$; Bounded antithesis $\in \mathsf{S} \cap \mathsf{C}$); C (Certificate acquisition, FA-REENTRY, both $\in \mathsf{L} \cap \mathsf{C}$; Bounded antithesis $\in \mathsf{S} \cap \mathsf{C}$); L (Certificate acquisition, FA-REENTRY $\in \mathsf{L} \cap \mathsf{C}$; Recurrence, Persistence, Response-after-recognition $\in \mathsf{L} \cap \mathsf{K}$); K (Integrity preservation, Never-externalise-without-condition $\in \mathsf{S} \cap \mathsf{K}$; Recurrence, Persistence, Response-after-recognition $\in \mathsf{L} \cap \mathsf{K}$). All membership vectors are assigned using the definitions of [Bra26a] Def. 2–5 (Safety = FIN-INV, Cosafety = FIN-CONF, Liveness = ¬FIN-INV i.e. $\mathrm{Bad}(\Phi) = \emptyset$, Coliveness = ¬FIN-CONF i.e. $\mathrm{Good}(\Phi) = \emptyset$); consequently $\mathbf{G}\,p$- patterns admit no good prefix and land in K, while $\mathbf{F}\,p$-patterns admit no bad prefix and land in L. This is the point at which engineering shorthand ("$\mathbf{G}\,p$ is a safety property", "$\mathbf{F}\,p$ is a liveness/reachability property") and the formal indicator vector χ disagree: informally $\mathbf{G}\,p \in \mathsf{S}$ and $\mathbf{F}\,p \in \mathsf{L}$ are correct, but the full cell is $\mathsf{S} \cap \mathsf{K}$ respectively $\mathsf{C} \cap \mathsf{L}$, because no finite good prefix ever permanently certifies $\mathbf{G}\,p$ and no finite bad prefix ever permanently refutes $\mathbf{F}\,p$. The minimum verdict rung follows [Bra26a] §10's ladder, understood as the smallest verdict-alphabet cardinality n sufficient to distinguish the cell: $\mathsf{S} \cap \mathsf{C}$ (clopen) is resolvable at $n = 2$; the one-sidedly monitorable cells $\mathsf{C} \cap \mathsf{L}$ (Cosafety ∩ Liveness) and $\mathsf{S} \cap \mathsf{K}$ (Safety ∩ Coliveness) admit a Kleene 3-valued monitor at $n = 3$ (the $\mathsf{C} \cap \mathsf{L}$ side emits $\top$ on a finite good prefix and ? otherwise, no $\bot$ ever needed; the $\mathsf{S} \cap \mathsf{K}$ side emits $\bot$ on a finite bad prefix and ? otherwise, no $\top$ ever needed); the two-sidedly non-monitorable $\mathsf{L} \cap \mathsf{K}$ cell (with the persistence Σ_2^0 vs recurrence Π_2^0 bipartition) requires the RV-LTL 4-valued ladder $n = 4$.

Remark 13.4 (Optional refinement-confirmation certification for FG-persistence entries). The verdict-alphabet ladder above operates on a fixed monitor against a temporal property Φ. A distinct and orthogonal certification layer is available when the monitor is embedded in a *refinement system* $\mathcal{R} = (D, A_0, U, E)$ in the sense of [Bra26c], with error metric E satisfying a Lyapunov strict-decrease property with constant $c > 0$ relative to an admitted trace class $\mathcal{T}$, and with a bounded initial error $E(A_0) \le E_{\max}$. Under those hypotheses, Theorem 1 of [Bra26c] yields a witnessing prefix of length at most $\lceil E_{\max}/c \rceil$ certifying that Φ holds relative to $\mathcal{T}$, and, with absorption on $C(A) \Leftrightarrow E(A) = 0$, the persistence upgrade $\tau \models \mathbf{F}\,\mathbf{G}\;C$ from [Bra26c] Proposition 2 (ii). Within the catalogue above this layer applies naturally to the FG-persistence row (Eventual persistence, $\mathsf{L} \cap \mathsf{K}$ persistence-Σ_2^0), where ordinary finite monitorability does not suffice, and only conditionally to the response family under a system-level refinement reading; on Cosafety rows (Certificate acquisition, Bounded antithesis, FA-REENTRY) the layer is redundant because a first good prefix already exists, and on the recurrence row ($\mathbf{G}\,\mathbf{F}$) it does not apply — [Bra26c] Proposition 2 (iii) explicitly denies $\mathbf{G}\,\mathbf{F}\;C$ from finite descent alone (a re-entry hypothesis is required). The layer is optional: it is invoked only when a Lyapunov refinement structure is present, and it certifies the persistence form $\mathbf{F}\,\mathbf{G}$, not the recurrence form $\mathbf{G}\,\mathbf{F}$.

Remark 13.5 (Failure modes as an architectural diagnostic). A joint failure to form an appropriate response can be located by programme layer: F-TYPE (required participant/property attribution not well-typed — *Critical Properties layer*), F-TEMP (temporal formulation Φ_p of the state predicate p has an $\mathsf{S}/\mathsf{C}/\mathsf{L}/\mathsf{K}$ signature $\chi(\Phi_p)$ that requires a verdict-alphabet rung the current monitor does not support — *`ltl_classes` layer* [Bra26a]), F-OBS/F-INFO (available observation model or the induced projection loses distinguishing information — *observation + LTL Triangles layer*; the same external world can lead to different monitor

verdicts under different observation models, as formalised by [Bra26d]), F-REASON (distinguishing information exists but the policy or evaluator that would use it is unavailable — *internal policy layer*), and F-ACT (response selected but externalisation carrier is unavailable — *Five Actions layer*). This is architectural diagnostic vocabulary for the present composition; we do not present it as a new general fault-isolation taxonomy. The corpus results locate specific failures at F-INFO (Proposition 9.6) and at F-REASON (Theorem 12.3); F-TYPE and F-ACT are addressed by the interface frameworks and are not proved by this paper. This is an architectural diagnostic, not a novel fault-isolation theory.

The LTL/Berne theorem core (Section 9, Section 12) remains formally self-contained. Critical Properties and Five Actions are supporting architecture and are cited via bibliography, not imported as formal dependencies.

14. Relation to DES, RV, and reactive synthesis

The mathematical machinery of Section 5 imports established DES and RV concepts. We enumerate them and explicitly disavow novelty for the imported apparatus.

Diagnosability. t_{obs} is a game-identification specialisation of the diagnosability delay of Sampath, Sengupta, Lafortune, Sinnamohideen, and Teneketzis [SSL$^+$95]. Their setting formalises detection of faults with finite delay under a partial-observation mask; our setting substitutes "faults" with "membership in a game family". The complexity of the sensor-selection problem defining the observation mask O is analysed by Yoo and Lafortune [YL02]; the timed and codiagnosable generalisations by Cassez [Cas12] could apply if our model is extended to timed traces.

Prognosability. t_{redir} is the intervention-window analogue of the reaction bound for coprognosis of Kumar and Takai [KT10], generalised from a fixed halt intervention to an arbitrary intervention class $\mathcal{I}$. Kumar and Takai's paper proves (§VIII) that diagnosability and prognosability are incomparable — the property that gives rise to the $W < 0$ regime we exhibit at NIGYSOB.

Runtime verification. The three-valued verdict semantics under which our O-observations resolve to game-identity or game-refutation is the RV framework of Bauer, Leucker, and Schallhart [BLS11]; the monitorability refinement of Peled and Havelund [PH19] sharpens the class placement of the fluent formulas of Section 4.

Runtime enforcement. The intervention class $\mathcal{I}_{\text{pol}}$ is a Berne-instance of the enforcement-primitive vocabulary of Falcone, Fernàndez, and Mounier [FFM12] (edit-automata suppress + insert), specialising Schneider's earlier safety-automata framework [Sch00] and the edit-automata renewal class of Ligatti, Bauer, and Walker [LBW09].

Supervisory control. The upstream frame for $\mathcal{I}$ is Ramadge–Wonham supervisory control [RW87]: the event alphabet E_g splits into $E_g^{\text{controllable}} \cup E_g^{\text{uncontrollable}}$, and $\mathcal{I}$ operates by disabling elements of $E_g^{\text{controllable}}$.

Reactive synthesis. The existence question "does there exist a reactive π realising the antithesis under $O, \mathcal{I}$?" is a GR(1) synthesis problem in the sense of Piterman, Pnueli, and Sa'ar [PPS05]; Pnueli–Rosner complexity [PR89] bounds it.

Enforcement monitors and shields. The monitor register $\mathcal{M}$ of Definition 5.6 follows the enforcement-monitor tradition of Bauer–Leucker–Schallhart [BLS11], Ligatti–Bauer–Walker edit-automata [LBW09], and the shield synthesis of Bloem, Könighofer, Könighofer, and Wang [BKKW15]. For five of the seven corpus games the register is singleton and $\mathcal{M}$ collapses to the observer of Definition 5.1; for TBFDP and SCHLEMIEL it is a bounded counter. The construction is a small specialisation of the enforcement-monitor stack, not a new object.

Monitor-machine interface for downstream reuse. Definition 5.6's tuple exposes the five fields (INPUT $= \Sigma$, STATE $= M$, UPDATE $= \mu$, TRIGGER $= \mu \cdot \gamma$-guard, OUTPUT $= \gamma$) that a downstream monitor design can plug into without renegotiating Definitions 5.7 and 5.8. A sixth field HYSTERESIS-NEEDED is false for every game in the present corpus (single-crossing or singleton). This interface is exposed for future reuse; no such plug-in monitor is introduced here.

Active observation as a natural DES extension. All observers of Definition 5.1 are passive functions of the history. A natural extension is to split $E_g^c = E_g^{c,a} \uplus E_g^{c,\mathrm{obs}}$ so that the observer contributes controllable probe events probe $\in E_g^{c,\mathrm{obs}}$ with echoes echo^{r+1}; anti-oracle admissibility (Definition 5.2) then generalises to a non-interference condition on the probe protocol in the sense of Volpano-style opacity. This falls outside the current paper's scope and is deferred to future work.

Our novelty is not the machinery. We claim no new theorem in DES, RV, or synthesis. The novelty of this paper is the source-faithful application to the Berne game corpus and the specific findings that emerge from the domain projection: role-level under-determination (Proposition 9.1) and permutation-insufficiency (Proposition 9.6). Neither emerges from a pure DES analysis; each is a domain-specific structural fact revealed by the projection.

15. Discussion

15.1. **The reactive/pre-committed distinction.** Berne's per-game antitheses [Ber64] were prescribed in an era before formal control theory, yet they naturally split along the axis that the DES literature would formalise decades later. Games with $W > 0$ admit reactive antitheses; games with $W \leq 0$ admit only pre-committed policies. This alignment is a scientific observation, not a metaphor: it is a consequence of the temporal geometry of the games themselves, and it explains why some of Berne's antitheses read as "mid-game moves" (WDYB) while others read as "life orientations" (NIGYSOB). The bridge to Kumar–Takai's coprognosability formalism [KT10] identifies the DES-native language for this distinction.

15.2. **Role under-determination and the substrate.** Proposition 9.1 exhibits three canonical games with pointwise-identical role traces. At the granularity of $\{\mathsf{Vic}, \mathsf{Per}, \mathsf{Res}\}$ alone, the games are indistinguishable. Discrimination requires either the conversational content that Berne uses to name each game (WDYB is defined by "yes, but..."; IWFY by "if it weren't for you"; SWYMD by "see what you made me do") or the substrate content that `berne_drama_substrate.py` attaches to each game. This is a formal statement of a clinical observation: role-labelling alone does not identify the game.

15.3. **The Learner motif.** Neither Berne nor Karpman introduces a Learner role. Where the Learner motif is useful in analysing an interaction, we formalise it as an epistemic policy $\pi^{\text{learn}} = \text{Receive} \to \text{Evaluate} \to \text{Experiment} \to \text{Update}$, applied at the policy layer of an existing agent — not as a fourth role. Introducing a fourth role would break the S_3 combinatorics of the switch algebra (Section 6), which is an internal-model consistency check, not an argument that no fourth psychological role can exist. Source fidelity is the primary reason not to introduce Learner as a role.

15.4. **Empowerment is out of scope.** David Emerald's Empowerment Dynamic [Eme05] (Creator/Challenger/Coach) has been proposed as a positive counterpart to the Drama Triangle. We do not analyse it in this paper: its peer-reviewed academic literature is sparse, and any Drama-to-Empowerment formalisation is out of scope for the bridge contribution we make here.

15.5. **An operational-capability reading of defended properties.**

Status: *PROGRAMME INTERPRETATION with formally-typed scope. Not a proved result of this paper.*

A common reading of each game attributes to one participant a proposition that they are said to "defend". Any such attribution introduces a doxastic inference the formal apparatus does not license (Section 15.6). A type-cleaner reading is available. For the interaction mechanisms represented in the present corpus, three *operational capabilities*, each definable from existing objects of Sections 4 and 5, suffice to factor the formal distinctions used throughout this paper:

- **Distinguishability of alternatives** (DIST) — agent a's observer separates the actual history from the alternative class at some finite prefix. Formally, $\mathsf{DIST}(a, O, g, k)$ holds iff for every $g' \in \mathfrak{A}(g) \setminus \{g\}$ one has $O(h_g[0..k]) \neq O(h_{g'}[0..k])$; this is exactly the machinery of Definition 5.3 viewed as a capability of the observer channel. It is not, and is not intended to be, a claim about internal cognitive recognition — WDYB's recognition of the yes-but pattern under O_{conv} remains atomic at the current abstraction boundary.
- **Availability of alternative admissible policies** (CHOOSE) — the set of policies in a given intervention sub-class (Definition 5.5) that admissibly extend a history h (Definition 5.9) has cardinality greater than one. This is a formal statement about policy availability, not about psychological freedom or autonomy.
- **Causal influence of controllable actions on successor state** (INFL) — there exist two controllable events $u, v \in E_g^c$ emitted by agent a at state q such that $\delta_g(q, u) \neq \delta_g(q, v)$. This is a qualitative existence statement over Definition 4.1's transition structure. Empowerment [KPN05] generalises it to an information-theoretic measure of agent control; this paper uses only the qualitative form.

Each capability is a derived view of an existing formal object; no new primitive is introduced. Together they factor the corpus: DIST underwrites t_{obs}-based recognition ($\mathfrak{N}^{\Diamond}$ for WDYB and IWFY; register-driven $\mathfrak{N}^{E}$ for TBFDP and SCHLEMIEL); CHOOSE underwrites the intervention-class selection of Definition 9.5; INFL underwrites the redirection guarantee that makes t_{redir} finite. Cases where a capability is present but is *not* the antithesis driver (e.g. DIST in SWYMD, NIGYSOB, KICK_ME) are precisely those in which the minimal successful class is $\mathfrak{N}^{\pi}$ rather than $\mathfrak{N}^{\Diamond}$ or $\mathfrak{N}^{E}$.

Scope of the claim. These three capabilities are the smallest family required to factor the mechanisms formalised in the present seven-game corpus; they are not claimed to be the universal basic properties of an agent. Persistence is not required as an operational primitive by this finite-horizon corpus, and relational- viability primitives are outside AP_g. Under Proposition 9.1 and Corollary 9.3 the role-projected trace is lossy; a capability-profile difference (Remark 9.8) is one of several upstream mechanisms consistent with each observed trajectory.

15.6. **Boundaries of the formal scope.** The manuscript makes no doxastic inferential claim: the observation channel of Definition 5.1 does not carry the agent's grammar or identity criterion, and no result in Section 9 depends on assuming what the agent believes. The per-agent state $X_a^r = (\rho_a^r, m_a^r)$ of Definition 5.7 is deliberately thin: it carries the role slot and the monitor register, and nothing else. Richer internal-agent architectures — carrying memory of past interactions, explicit models of other participants, action-polarity attribution against declared temporal targets, or accumulated incentive state — could distinguish dimensions that the Karpman projection π_K erases (Corollary 9.3); such architectures are outside the present formal scope and are the subject of separate programme work.

16. Limitations

(1) **Transactional Analysis is the source domain, not empirically validated by this paper.** We do not claim clinical validity for TA itself. TA has undergone independent psychometric validation of the Drama Triangle roles [LD20] and applied use [Nel14], but these are domain-external references, not results of the present work.
(2) **Karpman role labels are project projections.** Karpman 1968 [Kar68] does not analyse any of the seven Berne games formalised here. All V/P/R labellings in this paper are our formal projections of Berne's transactional structure onto Karpman's later role vocabulary.
(3) **Source-provenance disclosure.** The Antithesis paragraphs of WDYB, NIGYSOB, and SWYMD are cited from [Ber64] with primary content verified against the source; exact printed page anchors against the Grove Press 1964 first printing are still to be triangulated against a physical copy. The IWFY antithesis is cited via secondary corroboration only. The KICK_ME formal precommitment is explicitly a programme reconstruction (see PRE-COMMITTED-class discussion in Section 8), not a verbatim Berne antithesis. Per-passage provenance detail is tabulated in Appendix A.
(4) **Formula G is a Berne 1972 construct.** Its attribution to Berne 1972 (not Berne 1964) is corroborated by Karpman's own verbatim 2019 attestation [Kar19], cited in Section 2; the acronym-string form ($C + G = R \to S \to X \to P$) does not appear in [Ber64].

(5) **Corpus results, not universal theorems.** Propositions 9.1 and 9.6 are statements over the seven games modelled. They do not generalise universally without further work.
(6) W **is not a novel formal-methods invariant.** Remark 5.12. The $W < 0$ impossibility is folklore in DES engineering.
(7) **Machine coverage gaps.** SWYMD and SCHLEMIEL are absent from some auxiliary artefact grids. This does not affect the scientific results of Section 9.
(8) **Payoff model has no receiver-reaction channel** for the canonical games (assumption A1). Consequences for gratitude analysis are discussed in Section 15.
(9) **Two-party scope.** The paper's corpus is uniformly two-party plus observer; multi-party coordinated interventions, and any E_g^u-populated environment semantics, are out of scope and left for future work.
(10) **Karpman Victim role $\neq$ actual victimisation.** Section 16.1 states this explicitly. The formal theory does not imply that a person exposed to real violence, coercion, or danger should reinterpret rather than protect themselves. Interventions in this paper are formal interaction policies; they are not clinical recommendations.
(11) **Attention operator is absent.** A rigorous analysis of attention as a distinct observation-time operator is deferred to future work; see Section 4.

16.1. **Safety boundary.** The Karpman Victim role Vic is a modelling construct in a formal interaction system. It is not synonymous with actual victimisation. No result in this paper implies that a person exposed to actual violence, coercion, or material danger should adopt a formal role reinterpretation in preference to genuine protection. The intervention class $\mathcal{I}_{\text{pol}}$ is a formal policy over a symbolic interaction; it is not a clinical or personal-safety recommendation.

17. Conclusion

We have formalised a source-verified corpus of six canonical Berne transactional games plus one project-derived variant, imported the diagnosability–prognosability–enforcement machinery of DES/RV, and applied it to derive per-game antithesis-window signatures. Two domain-specific findings emerge: role traces under-determine the Bernian game within the corpus; intervention sub-type is not a function of the induced Karpman permutation alone. The formal-methods machinery is not novel; the contribution is the source-faithful bridge.

Future work includes closing the coverage gaps for SWYMD and Schlemiel in the auxiliary machine-checked grids; formalising an attention operator and re-examining t_{obs} under attention-varying observation models; extending the corpus to multi-party games with a formal uncontrollable-event apparatus; and — if primary-source access to Emerald 2005 is obtained — attempting a symmetric analysis of the Empowerment Dynamic.

Appendix A. Source provenance ledger

This ledger uses the following provenance scheme, defined here as the paper's own scholarly notation. Provenance levels: **P0** = primary text located and cited verbatim with a printed page anchor; **P0*** = primary text content verified against the cited work but exact printed page anchor pending against the specific Grove Press 1964 first printing; **S1** = secondary

attestation only; **S2** = triangulated via review literature. Status: ✓ = verified; † = content verified, printed page anchor pending; ∘ = content still to be verified against the primary source; △ = programme reconstruction.

Cited claim	Source	Level	Status
WDYB originated in Berne's work	[Ber58] (Hyacinth, pp. 740–741)	P0	✓
SCHLEMIEL antithesis (verbatim)	[Ber58] (p. 740)	P0	✓
Berne's game definition	[Ber64]	S1	✓
"Minimum number of moves . . . protocol"	[Ber64], opening chapter	P0*	†
Antithesis for WDYB (content-verified)	[Ber64], Antithesis paragraph	P0*	†
Antithesis for NIGYSOB ("correct behaviour")	[Ber64], Antithesis paragraph	P0*	†
Antithesis for SWYMD (three degrees)	[Ber64], Antithesis paragraph	P0*	†
Antithesis for IWFY	[Ber64], Antithesis paragraph	S1	∘
KICK_ME mechanism (provocation until reaction)	[Ber64], discussion body	P0*	†
KICK_ME formal precommitment countermove	programme reconstruction	—	△
Formula G is a Berne 1972 construct	[Ber72] + [Kar19] verbatim attestation	P0	✓
Drama triangle: three action roles	[Kar68]	P0	✓
"Drama = switches in role and location on a time continuum"	[Kar68] (verbatim)	P0	✓
"Games have one major switch"	[Kar68] (verbatim)	P0	✓
Diagnosability delay	[SSL+95]	P0	✓
Reaction bound for coprognosis	[KT10]	P0	✓
Runtime enforcement primitives	[FFM12]	P0	✓
Supervisor control alphabet split	[RW87]	P0	✓
Monitorability three-valued semantics	[BLS11]	P0	✓
GR(1) reactive synthesis	[PPS05]	P0	✓

Rows marked † (P0*) are content-verified against the primary source [Ber64] with Grove first-edition (192 pp.) page anchor pending final triangulation; rows marked ∘ remain S1-attested only. Neither status is blocking for the results of Section 9, which reason from role traces (Berne 1958 anchor for WDYB and SCHLEMIEL) and from Karpman 1968's P0 verbatim attestations for role structure and switch semantics.

Appendix B. Complete traces

Each row is the fluent sequence extracted from the accompanying software artefact (Appendix E) for that game. Format: $(q_0, \mathsf{viol}_0, \mathsf{payoff}_0, \mathsf{switch}_0) \to (q_1, \ldots) \to \cdots$. Details omitted; see the corpus of Section 7 for per-game antithesis-window computations.

Appendix C. Proof sketches

Proposition 9.1: direct trace comparison over the corpus of Section 7.

Proposition 9.6: exhibition of two distinct antitheses (WDYB Others cascade vs SWYMD Other policy invariant) for games sharing $\sigma_g = (\mathsf{Vic\ Per\ Res})$.

Appendix D. Exhaustive corpus tables

Tables 1, 2 and 4 in the main text summarise the corpus formalisation; a per-game per-observer t_{obs} / t_{redir} / W grid across the four observer profiles (O_{ext}, O_{conv}, O_a, O_∞) is included in the accompanying software artefact (Appendix E).

Appendix E. Artifact availability

All computational results in this manuscript are reproducible from an accompanying software artefact. The artefact contains: (a) the seven-game corpus formalisation with role traces and switch-permutation verification; (b) the observer projections $O_{\text{ext}}, O_{\text{conv}}, O_a, O_\infty$; (c) the t_{obs} / t_{redir} / W computation of Table 2 and its per-observer extension; (d) implementations of the intervention sub-types $\mathfrak{N}^\pi, \mathfrak{N}^\Diamond, \mathfrak{N}^E$ and the monitor machine $\mathcal{M}$ of Definition 5.6 for the counter-carrying games (TBFDP, SCHLEMIEL); (e) a regression test suite covering anti-oracle admissibility, per-game W values, S_3 generation, and the WDYB / IWFY / SWYMD role-trace equality of Proposition 9.1. On acceptance the artefact will be deposited to a public archive under a CC-BY 4.0 licence; DOI will be assigned at that time.

Implementation mapping (definition → artefact module): $\mathcal{G}_g$ (Definition 4.1) → corpus module; fluents (Definition 4.3) → property-classes module; σ_g (Definition 6.1) → switch-algebra module; Proposition 6.2 → switch-algebra regression tests; P_{lock} → property-classes module; $\mathcal{M}$ (Definition 5.6) → property-game module (TBFDP evidence accumulator; SCHLEMIEL forgiveness-counter).

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